When Do Foundation Models Pay Off for Retail Demand Forecasting?
A Systematic Review and Cross-Benchmark Meta-Analysis

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Abstract

We present a PRISMA-compliant systematic review of forecasting methods for short-term retail demand, combined with original gap-filling experiments that address structural holes in the literature. We screen approximately 150 papers (2020–2026) and extract quantitative results into 185 rows spanning M5, Favorita, and Rohlik v2. Because no existing paper reports both foundation models and gradient-boosted trees on these retail datasets under matched evaluation conditions, we run our own experiments (Source B: 45 rows from a consumer MacBook and Azure ML batch sweep) to enable cross-family comparison. A cross-paper pooled random-effects model (`metafor::rma.mv`, REML, Knapp–Hartung) on $k = 10$ bucket-level deltas finds $\hat{\Delta}(\mathrm{FM} - \mathrm{ML\_TREE}) = -0.044$ WAPE (95% CI $[-0.11, 0.03]$, $p = 0.17$) after excluding the M5 dataset, whose per-series WAPE metric produces a large but artefactual FM advantage driven by zero-denominator explosions in the LightGBM baseline. The M5 WAPE cells ($\Delta \approx -0.54$ to -0.78) and M5 WRMSSE cell ($\Delta = +0.41$, FM worse) are reported as a dataset-specific finding, not a family-level effect. We provide a practitioner-oriented cost–accuracy Pareto analysis and document that the LightGBM direct-vs-recursive protocol gap is conditional on demand intermittency, a finding vindicated by the gap-filling experiments.

Keywords: demand forecasting, foundation models, meta-analysis, LightGBM, systematic review, retail, time series

1. Introduction

The 2024–2026 wave of time-series foundation models — Ansari et al. (2025), TimesFM 2.5 (Das et al., 2025), Moirai 2.0 (?), TiRex (?) — has shifted the default assumption in short-term forecasting. Where a retail team in 2022 would start with LightGBM and a feature-engineering sprint, the 2026 playbook increasingly says: try a zero-shot foundation model first, tune LightGBM only if the FM disappoints. Each model paper reports favourable results on its own evaluation suite — GIFT-Eval, fev-bench, or a curated subset of the M competitions — but no paper runs the same battery of models under matched conditions

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on the same retail datasets and reports the delta. The benchmarks multiply; the synthesis does not.

This paper asks: **when do foundation models actually pay off for retail demand forecasting?** We operationalise "pay off" as a statistically significant reduction in per-series WAPE relative to a well-tuned, covariate-aware LightGBM baseline under a matched rolling-origin protocol. By "when" we mean: on which datasets, horizons, demand distributions, and hardware budgets.

1.1. Research questions

We address four research questions across three complementary lenses (accuracy, moderators, and cost):

- **RQ1 (Accuracy):** Do zero-shot foundation models outperform gradient-boosted tree baselines on retail WAPE? (§6.7)
- **RQ2 (Moderators):** Which data characteristics — demand intermittency, covariate richness, forecast horizon, multi-step protocol — moderate the FM-vs-ML_TREE gap? (§6.8)
- **RQ3 (Cost):** What is the cost-accuracy Pareto frontier when consumer-hardware electricity, cloud-GPU list price, and CO_2 footprint are included? (§6.5)
- **RQ4 (Guidance):** Under what conditions should a retail forecasting team deploy a foundation model instead of investing in feature engineering for LightGBM? (§8)

1.2. Approach

This study combines a PRISMA-compliant systematic review with original gap-filling experiments. The two-source design is not a methodological refinement — it is structurally necessary: as §4 documents, **no paper in the existing literature reports both foundation models and gradient-boosted trees on the same retail dataset under matched evaluation conditions.** Without our own experiments, the FM-vs-ML_TREE comparison on retail data is simply not possible from the literature alone.

1. **Source A (literature extraction).** We screen ~150 papers (2020–2026) from Scopus, Google Scholar, Semantic Scholar, and arXiv, applying retail-domain inclusion criteria (>1,000 series, quantitative accuracy metric on at least one of M5, Favorita, Rohlik v2, GIFT-Eval retail tasks, fev-bench retail tasks, or Walmart). The extraction yields 185 rows in a structured CSV with 23 fields per row (§3.2). Source A provides the ML_TREE and statistical baselines for the cross-paper pool and contextualises our FM results against the published literature.

2. **Source B (gap-filling experiments).** We run our own experiments under a unified evaluation protocol (§5.1): two foundation models (Chronos-Bolt-Tiny, TiRex) on a consumer MacBook (Apple silicon MPS), and three baselines (lightgbm_cov, lightgbm_direct, seasonal_naive) on an Azure ML batch cluster. Source B produces 45 rows with runtime, cost, and CO_2 metadata, keyed to Source A via a shared `paper_id` scheme that enables within-paper and cross-paper pairing (§6.7). Source B provides the primary FM retail cells that no existing paper covers.

The cross-paper analysis (§6) pools Source A and Source B into a random-effects model: for each (dataset, horizon, metric) bucket, we compute mean(FM rows)−mean(ML_TREE rows) as a single bucket-level delta, then fit `rma.mv` (REML, Knapp–Hartung) with $k = 10$ buckets clustered on dataset. **The FM side of this comparison is dominated by Source B** (our own experiments); Source A contributes the bulk of the ML_TREE side. This asymmetry is inherent to the current state of the literature and is a key finding in itself (§4). We report the pooled estimate with and without Source B to make the contribution of each source transparent.

1.3. Headline findings

Three findings motivate the sections that follow:

1. **The FM advantage is null on smooth-demand retail data.** Excluding M5, the cross-paper pooled delta is $\hat{\Delta} = -0.044$ WAPE (95 % CI $[-0.11, 0.03]$, $p = 0.17$). On Favorita, LightGBM with covariates matches Chronos-Bolt-Tiny within 0.3 pp at $h = 7$ — a marginal LGBM win. On Rohlik, FMs lead by 4–8 pp but the gap narrows with horizon and does not reach statistical significance in the meta-regression. The heterogeneity across the six excl-M5 buckets collapses to near-zero (Q $p = 0.85$), meaning the null result is not masking opposing effects.

2. **M5 is a dataset-specific finding, not a family-level effect.** The M5 WAPE cells show FM advantages of 54–78 pp over lightgbm_cov, but this is driven by a per-series WAPE denominator pathology on intermittent demand: several Azure lightgbm_cov runs logged WAPE_mean = ∞ (zero-denominator explosion on tail series). On the same M5 data using WRMSSE (the M5 competition metric), the sign reverses: competition-grade LightGBM scores 0.52 WRMSSE vs Chronos at 0.97. **The M5 WAPE and WRMSSE cells should not be averaged into a single "M5 effect."** The paper reports them as separate rows in Table 19 with an explicit metric caveat.

3. **The LightGBM protocol gap is conditional on intermittency.** §5.3 shows that the direct-vs-recursive multi-step gap flips sign at the continuous-demand boundary: direct wins on M5 (intermittent) by 17–22 %, recursive wins on Favorita and Rohlik (smooth) by 3–10 %. This conditional pattern, confirmed by Source B gap-filling, means that no retail-benchmark protocol recommendation that ignores demand distribution can survive cross-dataset replication.

1.4. Contributions

1. **First PRISMA-compliant systematic review** of foundation models for retail demand forecasting (185 extraction rows, 2020–2026).

2. **Gap-filling experiments** (45 rows) under a unified rolling-origin protocol with consumer-HW and cloud cost metadata.

3. **Cross-paper pooled meta-regression** identifying a null FM-vs-ML_TREE effect on smooth-demand retail and a metric-dependent M5 artefact.

4. **Baseline reliability analysis** documenting the conditional direct-vs-recursive LightGBM gap across three datasets.

5. **Practitioner decision framework** mapping data characteristics to model-family recommendations with cost-accuracy Pareto figures.

1.5. Paper structure

§2 surveys the forecasting taxonomy and existing benchmarks. §3 describes the PRISMA search protocol, extraction schema, and gap-filling experimental design. §4 reports the literature extraction results (PRISMA flow, descriptive statistics). §5 presents the gap-filling experiments: §5.1–6.3 experimental setup and per-dataset results, §5.3 baseline reliability (direct vs recursive), §5.4.3 foundation model consumer-box results. §6 runs the meta-regression: cross-paper pooled deltas (§6.7), moderator hypotheses (§6.2–6.8), cost-accuracy Pareto frontier (§6.5), heterogeneity diagnostics (§6.6), and preliminary findings (§6.7). §8 translates the findings into a practitioner decision framework. §8.4 discusses limitations ($k = 10$ pool, metric sensitivity, consumer-HW-only Source B). §9 concludes with per-RQ answers and future work.

2. Background

Short-term retail demand forecasting — predicting daily or weekly SKU-level sales 1–4 weeks ahead — is one of the most mature applied forecasting problems. The operational stakes (replenishment, markdown optimisation, labour scheduling) have driven a long lineage of benchmark competitions and model development. This section establishes the taxonomy we use throughout the paper and surveys the three benchmark ecosystems whose results populate our extraction schema.

2.1. Taxonomy of forecasting approaches

We group methods into four families, matching the `model_family` field in our extraction schema (§3.2). The grouping is coarse by design: the meta-regression (§6) treats it as a categorical moderator, so finer distinctions (e.g., GRU vs Transformer within NN) would create sparsely populated cells.

Statistical models.. Exponential smoothing (ETS), ARIMA, Theta, Seasonal Naive, and intermittent-demand methods (Croston, TSB). These are fitted per-series with no cross-series learning. Their value in this paper is as a sanity-check baseline: any global model that loses to Seasonal Naive on a given (dataset, horizon) cell is failing in a way that demands explanation, not hand-waving. We include Croston because M5 has a ~70 % zero-day fraction and Croston is the textbook method for intermittent demand. In our extraction schema these models carry `model_family = "statistical"`.

Gradient-boosted trees (ML_ TREE).. LightGBM and XGBoost fitted as global models: one model trained across all series, with cross-sectional features (lags, calendar effects, price, promotions, store/item embeddings). This is the family that the M5 competition (§2.2) crowned as the winner, and it remains the production default at most retailers. In the extraction schema, `model_family ∈ {"ml_tree", "ml", "ml_gbm", "gbdt", "ml_ensemble",` `"ml_hybrid"}`. The §5.3 baseline reliability analysis shows that the choice of multi-step strategy (direct vs recursive) within this family can flip the accuracy ranking by 10+ pp WAPE depending on demand distribution — a confound that the literature rarely reports and that our gap-filling experiments control for.

Deep learning (NN).. Recurrent (DeepAR, LSTNet), attention-based (TFT, PatchTST, Informer), and MLP-based (N-BEATS, N-HiTS) models trained on many series. These models occupy a middle ground: they learn cross-series patterns like tree models but have higher computational cost and typically require more data to avoid overfitting. In the M5 competition, the top-50 solutions were overwhelmingly tree-based; DL entries placed in the top quartile but did not win outright. Since 2023, PatchTST and its descendants have narrowed the gap on benchmark suites. In the extraction schema, `model_family` $\in$ `{"nn", "deep", "transformer", "rnn", "deep_learning", "ensemble_dl"}`.

Foundation models (FM).. Pre-trained on large corpora of heterogeneous time series and applied zero-shot (no fine-tuning on the target dataset) or with lightweight fine-tuning. The models in scope for this review are:

Table 1: Foundation models in scope.

Model	Params	Architecture	Covariates	Key reference
Chronos / Chronos-2	9–710 M	Encoder-only T5	v2 only	Ansari et al.
TimesFM 2.5	200 M	Decoder-only	No	Das et al. (20
Moirai 2.0	11–305 M	Decoder-only (MoE)	Yes	**?**
TiRex	~35 M	xLSTM	No	**?**
TabPFN-TS	~12 M	Tabular PFN + temporal featurization	No	Müller et al.
Lag-Llama	6 M	Llama backbone	No	**?**
TimeGPT	undisclosed	Proprietary	Yes	Garza and M

In the extraction schema, `model_family = "foundation"`. The zero-shot setting is the primary scope of this review; fine-tuned FM entries (e.g., Chronos-2 with in-domain pre-training) are retained in the schema but flagged as `fine_tuned = "Yes"` and excluded from the primary meta-regression to avoid confounding pre-training scale with task-specific tuning.

2.2. Existing benchmarks

Three benchmark ecosystems contribute the majority of our extraction rows. We note their design choices because those choices determine what the extracted metrics actually measure — and where they systematically mislead.

M5 Competition (Makridakis et al., 2022).. 30,490 daily Walmart series across 10 stores and 3,049 products, with calendar, price, and SNAP covariates. The official metric is WRMSSE — a weighted ratio of root mean squared scaled errors that accounts for intermittent demand by scaling each series' error by its historical variability. The competition was won by gradient-boosted trees (top-5 all LightGBM or XGBoost); the best DL entry (DeepAR) placed around #50. M5 is still the go-to retail benchmark in 2026, but it has two ageing problems: (1) the data is from 2011–2016, predating modern promotional patterns, and (2) many foundation model papers report per-series WAPE rather than WRMSSE on M5, creating a metric mismatch that §5.4.4 documents in detail. On per-series WAPE, the ~70 % zero-day fraction in M5's long tail causes denominator explosions that make LGBM appear to perform worse than Seasonal Naive — a metric artefact, not a model failure.

GIFT-Eval (Aksu et al., 2024).. 28 datasets (144,879 series) curated to avoid train–test leakage in foundation-model evaluations. Covers multiple domains including retail, energy, and transport. The standard metric is a skill score relative to Seasonal Naive, computed with bootstrapped confidence intervals. GIFT-Eval is the dominant evaluation suite in the FM literature: most of our Category A (FM paper) rows cite it. Its limitation for our purposes is that only a subset of the 28 datasets is retail-specific, and the aggregate skill scores mix retail with non-retail, making it hard to extract a clean retail-only FM-vs-ML_TREE delta.

fev-bench (Shchur et al., 2025).. 100 tasks spanning 46 with exogenous covariates, designed as a direct response to the "no covariates in FM evaluation" criticism. Uses bootstrapped CIs and skill scores. Covers a broader set of model families than GIFT-Eval, including covariate-aware baselines that are largely absent from FM papers. For our extraction, fev-bench provides the largest single block of rows with both FM and statistical baselines on the same tasks, though the task-level granularity (rather than dataset-level) makes cross-benchmark alignment non-trivial.

2.3. Meta-analyses in forecasting

The M-competition lineage (M1–M5, Makridakis 1982–2022) is the closest antecedent to our work: each competition is, in effect, a large-scale empirical comparison of forecasting methods on a shared dataset with standardised metrics. The M5 (retail) and M4 (heterogeneous) competitions produced substantial secondary analysis, but always on a single dataset per competition. Petropoulos et al. (2022) survey the forecasting landscape broadly but do not run a PRISMA-compliant meta-regression. Lim & Zohren (2021) review deep learning for time series but predate the FM wave.

To our knowledge, **no prior work combines** (1) a PRISMA-compliant systematic search, (2) quantitative extraction across multiple benchmarks, (3) gap-filling experiments to fill structural holes, and (4) a random-effects meta-regression with moderators — for foundation models in the retail forecasting domain. This is the gap we fill.

3. Methodology

This section describes the three-phase protocol: search and extraction (Source A), gap-filling experiments (Source B), and the meta-regression specification that combines them.

3.1. Search protocol (PRISMA)

We follow the PRISMA 2020 reporting guidelines (Page et al., 2021) for systematic reviews. The full search protocol is committed at `analysis/prisma_search_protocol.md`; we summarise the key parameters here.

Databases. Scopus, Google Scholar, Semantic Scholar API, arXiv (cs.LG, stat.ML), and Web of Science.

Search query. We combine three concept blocks with AND:

1. *Foundation model identifiers:* "foundation model", "pretrained model", "zero-shot forecasting", Chronos, TimesFM, Moirai, TimeGPT, Lag-Llama, TiRex, TabPFN.

2. *Demand forecasting:* "demand forecasting", "retail forecasting", "time series forecasting", "sales forecasting".

3. *Benchmark anchors:* "benchmark", "comparison", "evaluation", M5, GIFT-Eval, fev-bench.

Date range. 2020-01-01 to 2026-04-12. The lower bound captures the M5 competition and pre-FM baselines; the upper bound is the extraction lock date.
Inclusion criteria.

1. Empirical evaluation of at least one foundation model on demand or retail forecasting.

2. Reports quantitative accuracy metrics (WAPE, MASE, MAE, sMAPE, CRPS, WRMSSE, or win-rate/skill-score equivalents).

3. Uses at least one of M5, Favorita, Rohlik v2, GIFT-Eval (retail tasks), fev-bench (retail tasks), or Walmart.

4. Peer-reviewed or published preprint with reproducible methodology.

5. English language.

Exclusion criteria.

1. No quantitative results (position papers, surveys without new data).

2. Only financial, stock, or energy forecasting with no retail overlap.

3. FM paper without forecasting evaluation (architecture-only).

4. Duplicate results — keep most recent version.

5. Non-time-series forecasting (causal inference only).

3.2. Data extraction (Source A)

Each included study is extracted into one or more rows of the structured CSV `analysis/extraction_sc` (185 rows as of 2026-04-15). The schema has 23 fields per row:
Row-level IDs. Each extracted result is a separate row with a unique `paper_id`. A paper reporting Chronos and LightGBM on M5 contributes at least two rows (`A01_chronos_indomain`, `A01_chronos_zeroshot`, etc.). This scheme means that `group_by(paper_id)` in the meta-regression returns singleton groups for most rows — a structural property that motivated the cross-paper pooling design of §4.
Category prefixes. `A` = FM papers (17 papers, 58 rows), `B` = benchmark papers (4 papers, 40 rows), `C` = retail ML/DL papers (8 papers, 31 rows), `D` = cross-domain papers (3 papers, 3 rows), `F` = fev-bench entries (1 paper, 8 rows), `OWN_*` = our own baseline runs from §5.2–5.3 (6 experiment blocks, 45 rows).

Table 2: Extraction schema fields (Source A).

Field	Type	Description
`paper_id`	char	Unique row-level ID (e.g., `A01_chronos_indomain`)
`authors`	char	First author et al.
`year`	int	Publication year
`title`	char	Paper title
`venue`	char	Journal, conference, or "this study"
`dataset`	char	Raw dataset name as reported
`dataset_variant`	char	Sub-dataset or task identifier
`n_series`	char	Number of series (free text)
`series_length_median`	char	Median series length
`frequency`	char	Data frequency (D, W, M, mixed)
`has_covariates`	char	Yes / No / mixed
`model_name`	char	Model name as reported
`model_family`	char	Normalised family tag (§4)
`zero_shot`	char	Yes / No / mixed
`fine_tuned`	char	Yes / No
`horizon`	char	Forecast horizon (free text)
`eval_method`	char	rolling / rolling-tail / fixed_origin
`metric_name`	char	Metric name
`metric_value`	char	Reported value (free text, incl. "best")
`runtime_reported`	char	Whether runtime is reported
`gpu_hours`	char	GPU hours if reported
`hardware`	char	Hardware description
`notes`	char	Free-text provenance notes

Normalisation. Raw `dataset` values are collapsed to five canonical names (M5, Favorita, Rohlik, fev-bench, Walmart) by the R function `normalize_dataset()` in `analysis/meta_regression.R`. Raw `model_family` values are collapsed to four analysis families (FM, ML_TREE, STATS, NN) by `normalize_family()`. Rows that do not map to a canonical dataset or family are excluded from the primary meta-regression but retained in the schema for provenance.

3.3. Gap-filling experiments (Source B)

Source A has two structural gaps: (1) no FM paper reports per-series WAPE on M5, Favorita, or Rohlik under a rolling-origin protocol matching §5.1.3 — the FM literature evaluates on benchmark suites (GIFT-Eval, fev-bench), not on individual retail datasets. (2) No paper reports both FM and ML_TREE on the same retail dataset under matched conditions with a shared `paper_id`, so within-paper pairing (the original §6.1 Pass 1 design) is structurally empty on Source A.

Source B fills these gaps with 45 rows from two hardware backends:

Consumer MacBook (M_SERIES_MAC). Apple M3 Air, 16 GB unified memory, PyTorch MPS backend. Two FM models:

- **Chronos-Bolt-Tiny** (9 M params, encoder-only): 9/9 cells (3 datasets × 3 horizons), all on MPS.
- **TiRex** (∼35 M params, xLSTM): 9/9 cells. Seven completed on MPS; two (Rohlik $h = 14$, $h = 28$) required CPU fallback (`TIREX_FORCE_CPU=1`) due to a pathological MPS dispatch in `xlstm_kernels` that stalled at <3 min CPU per hour wall time.

Azure ML batch (AZURE_E4DS_V4). Standard_E4ds_v4 (4 vCPU, 32 GB), CPU-only. Three baseline models:

- **lightgbm_cov** (LightGBM recursive, Tweedie + 6 covariates): 9 cells.
- **lightgbm_direct** (LightGBM direct, Tweedie + 6 covariates, per-horizon head): 9 cells.
- **seasonal_naive**: 9 cells.

All 45 cells use the §5.1.3 evaluation protocol: rolling-origin with 100 sampled series per cell, per-series WAPE as the primary metric, and a fixed random seed for reproducibility. Runtime, cost, and CO_2 are logged per run via MLflow (local file store for MacBook runs, Azure ML workspace for batch runs). The export script `tools/export_mlflow_to_csv.py` pulls both backends into a single CSV keyed on the shared `paper_id = LOCAL_MAC_<dataset>_h<horizon>`, which is what enables within-paper Δ computation in the R pipeline.

3.4. Meta-regression specification

The primary estimand is the cross-paper pooled delta:

> For each (dataset, horizon, metric) bucket that contains at least one FM row and at least one ML_TREE row from any source (A or B), compute Δ = mean(FM metric values) – mean(ML_TREE metric values).

This produces $k = 10$ bucket-level deltas (Favorita $\times$ 3 horizons $\times$ WAPE, M5 $\times$ 3 horizons $\times$ WAPE, M5 $\times$ long $\times$ WRMSSE, Rohlik $\times$ 3 horizons $\times$ WAPE).

Notation convention. Throughout this paper, $\hat{\Delta}$ denotes the pooled estimate of the FM − ML_TREE difference on the metric in question (WAPE unless stated otherwise). **Negative $\hat{\Delta}$ means FM is better** (lower WAPE); **positive $\hat{\Delta}$ means ML_TREE is better.** All confidence intervals and p-values refer to the two-sided test $H_0\colon \Delta = 0$. When we write "the FM advantage is null," we mean the CI includes zero — not that $\hat{\Delta}$ is exactly zero. The variance proxy v_i is defined per bucket in the next paragraph; it is *not* the metric's own variance but a precision weight reflecting pool depth and dataset size.

The model is:

```
rma.mv(yi = delta, V = vi,
       random = ~ 1 | dataset_norm,
       data = delta_cross,
       test = "t",          # Knapp-Hartung
       method = "REML")
```

where v_i is the per-bucket sampling variance proxy defined above.

Random effect. `~ 1 | dataset_norm` clusters on dataset because each Δ mixes rows from multiple papers and the within-paper anchoring is lost by construction. The dataset level captures the primary source of heterogeneity (M5 vs smooth-demand datasets).

Variance. For the cross-paper pooled design, the per-bucket sampling variance is $v_i = (1/n_{\text{FM}} + 1/n_{\text{ML\_TREE}})/n_{\text{series,hm}}$, where n_{FM} and $n_{\text{ML\_TREE}}$ are the number of FM and ML_TREE rows entering the bucket and $n_{\text{series,hm}}$ is the harmonic mean of the per-row series counts within the bucket. This proxy weights buckets with more source rows and larger datasets more precisely than uniform variance. For within-paper paired Δ (Source B cells), $v_i = 1/n_{\text{valid}} = 0.01$ because each cell samples 100 series (§5.1.3). The proxy does not capture within-series error correlation or between-paper protocol heterogeneity; these are flagged as limitations in §8.4.

Small-sample adjustment. `test = "t"` invokes the Knapp–Hartung adjustment (Knapp and Hartung, 2003), replacing the standard normal reference distribution with a t-distribution whose degrees of freedom equal $k - p$ (where p is the number of estimated coefficients). This is critical at $k = 10$.

Moderator regressions. We test three moderators individually:

1. `mods = ~ dataset_norm` — tests whether the FM-vs-ML_TREE gap varies by dataset (H1 proxy for intermittency, since M5 is the only high-intermittency dataset in the pool).
2. `mods = ~ horizon_bucket` — tests whether longer horizons widen or narrow the gap.
3. `mods = ~ has_covariates` — tests whether covariate-rich ML_TREE baselines reduce FM advantage (H2).

Sensitivity analyses. Two pre-registered sensitivity runs:

- **Excl-M5:** Drops all M5 cells and refits on $k = 6$ (Favorita + Rohlik WAPE buckets only). This is the cleanest test of whether FMs outperform ML_TREE on smooth-demand retail.

- **Within-paper only (Source B):** Reverts to the original strict `group_by(paper_id, dataset_norm, horizon_bucket, metric_name)` design, which yields $k = 9$ cells all from `LOCAL_MAC_*`. This tests whether the cross-paper pooling design materially changes the conclusion.

Implementation. All fits are computed by `analysis/meta_regression.R` using the `metafor` R package (Viechtbauer, 2010). Outputs (forest plots, intercept tables, cell-level CSVs) are committed to `analysis/figures/`.

4. Literature Results

This section reports the outcome of the PRISMA search and describes the 185-row extraction before the gap-filling experiments (§5) and meta-regression (§6) operate on it.

4.1. PRISMA flow

[Figure 4.1: PRISMA flow diagram — see `analysis/figures/prisma_flow.pdf`, rendered from `analysis/prisma_flow_diagram.py`.]

Table 3: PRISMA flow summary.

Stage	Count	Notes
Records identified	∼150	Scopus + Google Scholar + Semantic Scholar + arXiv
Duplicates removed	∼30	Cross-database overlap, especially arXiv ↔ Scopus
Records screened (title/abstract)	∼120	
Excluded at screening	∼50	Non-retail, no quantitative results, financial only
Full-text assessed	∼70	
Excluded at full-text	∼32	No retail dataset, no FM evaluation, duplicate results
Studies included	38	32 external papers + 6 own experiment blocks
Extraction rows	185	Multiple rows per study (model × dataset × metric)

The 38 included studies break down as follows:

- **Category A (FM papers):** 17 papers, 58 rows. These are the primary FM technical reports — Chronos (A01), Chronos-2 (A02), TimesFM (A03–A04), Moirai (A05–A06), TiRex (A13), TabPFN-TS (A14), and the smaller FM entries (A07–A11, A15–A17).

- **Category B (benchmark papers):** 4 papers, 40 rows. GIFT-Eval (B01), fev-bench (B02), and two multi-model comparison papers (B03, B07–B08).

- **Category C (retail ML/DL papers):** 8 papers, 31 rows. M5 competition analyses (C01, C05), transformer-based retail studies (C03, C06), AutoML baselines (C10), and cross-domain comparisons (C04, C12).

- **Category D (cross-domain):** 3 papers, 3 rows. Included for model-family coverage but not in the retail-scoped meta-regression.
- **Category F (fev-bench entries):** 1 paper, 8 rows.
- **OWN_* (own experiments):** 6 experiment blocks, 45 rows. LightGBM baselines (§5.2–5.3) and Seasonal Naive across three datasets and three evaluation protocols.

4.2. Descriptive statistics

Temporal distribution. The extraction skews heavily toward 2024–2026: 79 rows from 2025 papers, 53 from 2026, 33 from 2024. Pre-2024 rows (20 total) are almost entirely M5 competition entries (2020–2022) and early DL comparisons. This reflects the recency of the FM wave — the first Chronos paper appeared in late 2023, and the bulk of FM evaluation literature dates from mid-2024 onward.

Dataset distribution. M5 dominates with 40 rows (22% of the extraction), followed by fev-bench (33 rows, 18%), GIFT-Eval (24 rows, 13%), and our three retail datasets (Favorita 10, Rohlik 10, Walmart 2). The remaining rows cover cross-domain or unnamed datasets. The concentration on M5 is partly an artefact of the M5 competition's longevity (every retail ML paper since 2020 benchmarks on it) and partly a genuine signal that M5 is the only public retail dataset with enough community coverage to populate a cross-paper comparison.

Model family distribution. Foundation models account for 76 rows (41%), followed by statistical (26 rows, 14%), ML/ML_TREE (25 + 5 = 30 rows, 16%), deep learning (13 rows, 7%), and various mixed/comparison/hybrid categories (40 rows, 22%). The high FM share reflects the search strategy's emphasis on FM papers (block 1 of the query); the OWN_* baseline rows partially offset the imbalance by contributing 45 ML_TREE + STATS rows.

Metric heterogeneity. The extraction contains 18 distinct metric names. The most common are WRMSSE (23 rows, mostly M5), WAPE (18 rows, mostly Source B and retail), win_rate_SQL (15 rows, fev-bench), MASE (13 rows), MAE_mean (9 rows), and avg_rank (8 rows). 14 rows carry `metric_name = "-"` (comparison or qualitative rows with no single numeric metric). This metric heterogeneity is a key challenge for the meta-regression: the cross-paper pooled Δ (§6.1) can only operate within a single metric scale, so each (dataset, horizon, metric) bucket must be metric-homogeneous. In practice, this means the primary regression runs on WAPE buckets ($k = 9$) and a single M5 long WRMSSE bucket ($k = 1$), for a total of $k = 10$.

Evaluation protocol. Three protocols appear in roughly equal proportion: rolling origin (46 rows), rolling-tail (39 rows), and fixed origin (38 rows). The remaining rows have no explicit protocol label. Rolling and rolling-tail are our matched Source B protocol (§5.1.3); fixed-origin rows from Source A may not be directly comparable but are included in the cross-paper pool where they share a (dataset, metric) bucket with rolling-origin rows. This is a known threat to validity (§5.4.8).

4.3. Narrative synthesis by model family

Foundation models. The 76 FM rows paint a consistent picture on benchmark suites: Chronos-2, TimesFM 2.5, and Moirai 2.0 achieve win rates of 75–84% against statistical baselines on GIFT-Eval, and skill scores of 20–35 on fev-bench. However, almost none of these rows report per-series WAPE on an individual retail dataset (M5, Favorita, Rohlik) under a rolling-origin protocol. Only two Source A FM rows land in the retail-scoped meta-regression: C05 (TEMPO baseline, WRMSSE 0.97 on M5 $h = 28$) and C03 (TimesFM on Rohlik, MASE+sMAPE with no numeric WAPE). This means that **the FM side of the retail meta-regression is almost entirely populated by Source B** — a structural limitation documented in §6.7.

Gradient-boosted trees (ML_TREE). The 30 ML_TREE rows split between M5 competition entries (C01, 7 rows with WRMSSE 0.52–0.54 for the top-5 solutions) and our own LightGBM baselines (OWN_*, 23 rows across three datasets with WAPE and WRMSSE). Source A ML_TREE rows are concentrated on M5 WRMSSE, where competition-grade LightGBM achieves 0.52 — materially better than any FM row on the same metric (Chronos at 0.97, from C05). This WRMSSE reading is the mirror image of the WAPE reading on the same dataset, where LGBM posts 1.35–1.94 (per-series) vs FM at 0.93–0.96 — the metric choice determines the sign of the FM-vs-ML_TREE comparison on M5.

Statistical models. The 26 STATS rows are mostly our own Seasonal Naive baselines (OWN_*, 20 rows) plus 1 Croston row from the M5 competition (C01) and a few MASE/sMAPE entries from cross-domain papers. Seasonal Naive provides the universal comparison floor: on Favorita and Rohlik, both FM and ML_TREE beat it comfortably (by 8–15 pp WAPE); on M5, Seasonal Naive at 1.31–1.33 WAPE actually beats lightgbm_cov (1.62–1.94) at longer horizons — a striking failure of the tree baseline documented in §5.3 and §5.4.5.

Deep learning. The 13 NN rows include M5 competition entries (DeepAR at WRMSSE 0.56 from C01, and PatchTST variants from C05) and transformer comparisons on M5 (C06). None of these rows share a (dataset, metric) bucket with FM rows in a way that enables direct FM-vs-NN comparison in the meta-regression. An FM-vs-NN analysis is possible on GIFT-Eval skill scores but is outside the retail scope of this paper.

4.4. Structural finding: no within-paper FM-vs-ML_TREE pairing in the retail literature

The single most important observation from the extraction is negative — and is itself a contribution of this review: **no Source A paper in the retail slice reports both FM and ML_TREE results under a shared paper identifier and a common per-series metric.** The FM papers (Category A) evaluate FMs against statistical baselines or against other FMs on benchmark suites, not against well-tuned LightGBM on individual retail datasets. The ML_TREE papers (Category C) predate the FM wave or focus on competition results without FM comparisons. The benchmark papers (Category B) include both families but on aggregate skill scores that cannot be decomposed to (dataset, horizon) cells.

This gap has two consequences. First, it is the reason Source B exists (§3.3): answering "do FMs beat LightGBM on retail data?" requires original experiments because the literature does not contain the comparison. Second, it means the cross-paper pooled analysis (§6.1) is a **systematic review augmented with original experiments**, not a classical meta-analysis of homogeneous effect sizes. The FM side of the retail comparison is populated almost entirely by Source B; the ML_TREE side draws on both Source A (competition-grade results) and Source B (our own baselines). This asymmetry is inherent to the current state of the FM literature — it is the gap our study fills, not a design flaw. We report the pooled estimate with and without Source B rows in §6.7 so readers can assess each source's contribution.

5. Gap-Filling Experiments

5.1. Experimental Setup

All gap-filling experiments described in §5 follow a single experimental protocol, parameterized per dataset. This section defines the protocol once. Dataset-specific choices (covariate sets, eval-window lengths, series-count caps) are noted inline in §5.2 for baseline models and §5.4 for foundation models.

5.1.1. Datasets

We evaluate on three retail demand forecasting datasets, chosen to span the distribution of zero-day fractions in public retail benchmarks while keeping the hierarchical-SKU × location structure that retail practitioners actually face. The three datasets together cover ∼211k × 1.4k series-days across the full product × location × time tensor.

Table 4: Datasets used in the gap-filling experiments. `Zero-day fraction` is the fraction of series-day observations with $y = 0$ in the training portion, a core moderator variable in §6.

Dataset	n_{series} (full)	n_{series} (used)	n_{days}	Frequency	Zero-day frac	Covariates used
M5	30,490	30,490	1,941	daily	∼70 %	`sell_price`, `snap_CA/TX/W` `is_event`, `wday`, `month`
Rohlik v2	5,390	5,390	1,402	daily	∼1.2 %	`sell_price_main`, `holiday`, `shops_close` `winter_school_holidays`, `school_holidays`, `dayofwee` `month`
Favorita	∼174,685	30,000	1,684	daily	∼15–25 %, long tail	`onpromotion`, `dcoilwtic` `is_holiday`, `transaction` `dayofweek`, `month`

Three notes on dataset preparation:

- **M5** is used in full. The 30,490 store-item series correspond to the Walmart subset of the M5 competition (Makridakis et al., 2022); we use the official tail 389 days as the evaluation window (§5.1.3).
- **Rohlik v2** is used in full. The dataset is 5,390 warehouse-SKU series from the Kaggle competition `rohlik-sales-forecasting-challenge-v2`, covering 2020-08-01 to 2024-06-02 across 7 warehouses (Prague ×3, Brno, Munich, Frankfurt, Budapest). Approximately 31% of Rohlik SKUs enter later than day 1 (ragged series starts), which we handle at the wide-pivot step via `fillna(0)` — see §5.1.4.
- **Favorita** is capped at the top 30,000 series by total `unit_sales` across the training portion, matching M5's scale for cross-dataset forecast-count comparability. The full

dataset has $\sim$174,685 series at $\sim$125 M rows and does not fit the 32 GB E4DS_V4 compute used for Phases A–D. The selection rule biases Favorita LGBM evaluation toward higher-velocity series; this is flagged as a limitation in §5.3.7 and §7, and the full-series evaluation is left for future work on larger hardware.

All three datasets are registered as versioned Azure ML data assets (`azureml:m5-sales:1`, `azureml:rohlik-train-v2:1`, `azureml:favorita-train:1`, plus per-dataset covariate assets) for exact reproducibility.

5.1.2. Compute and Tracking

All jobs run on Azure ML compute cluster `cc-forecast-batch` (E4DS_V4 VM, 4 vCPU / 32 GB RAM / no GPU) in region `swedencentral`, inside Docker environment `forecast-benchmark-cpu-env:1`. We use `cc-forecast-batch` for the classical and ML baselines of §5.2–§5.3. Foundation model sweeps (§5.4) run on consumer hardware (Apple M-series MPS); setup details are in §5.4.1.

Experiment orchestration uses Azure ML pipelines defined in `benchmark/code/pipelines/{m5,rohlik` Every run logs metrics, parameters, runtime, and per-series CSVs via MLflow to the workspace tracking server. Cost and CO_2 estimates are computed inside the job using Azure's published per-second pricing for `Standard_E4ds_v4` in `swedencentral` ($0.38/hr at time of writing) and the region's marginal CO_2 intensity (0.041 kgCO_2eq/kWh, swedencentral 2024 baseline; Nowtricity/Ember). The per-job runtime, cost, and CO_2 columns in Tables 7–10 are computed this way and are therefore comparable across datasets.

5.1.3. Evaluation Protocol

For each dataset we use **rolling-origin non-overlapping evaluation with a held-out tail window**, which is the protocol most commonly used in the FM papers we screened (§4) and which gives a clean, reviewer-auditable split. Specifically:

- **Train / eval split:** `train_until` $= \lfloor 0.8 \times n_{\text{days}} \rfloor$. This gives a tail-eval window of 389 days on M5, 281 days on Rohlik, and 337 days on Favorita. The choice of 0.8 follows **?** and Shchur et al. (2025).

- **Rolling origin:** within the tail window, forecast origins are spaced exactly h days apart so that windows are non-overlapping. For example at $h = 14$ on Rohlik (eval window = 281 days), we have $\lfloor 281/14 \rfloor = 20$ evaluation windows per series. This keeps per-horizon sample counts independent of each other and avoids the optimistic bias of overlapping-windows evaluation.

- **Horizons:** $h \in \{7, 14, 28\}$ on all three datasets. Seven days is the operational horizon most retailers use for short-term replenishment; 28 days is the M5 competition horizon and a common upper bound in the literature (Makridakis et al., 2022); 14 days bridges the two. We report every metric at every horizon to enable horizon-conditioned interpretation.

- **Metrics:** MAE, sMAPE, WAPE, and (on M5 only) WRMSSE. MAE and WAPE are computed per series and averaged with equal weights (`_mean` suffix in the tables).

sMAPE uses the classical symmetric form with a 200% cap on zero actuals. WRMSSE is computed in-job via the full 12-level Makridakis et al. (2022) hierarchy from the raw sales/calendar/prices CSVs. On retail tasks with heavy right tails (all three of ours), sMAPE is reported only for reproducibility and is not used in any primary comparison — see §5.3.9 for the cross-dataset justification.

- **Per-series filtering for WAPE:** WAPE requires a non-zero denominator (sum of actuals in the eval window). For series where the denominator is zero we drop the series from WAPE computation and report the `n_valid` count alongside the metric in Tables 7–10. This affects at most 0.5 % of M5 series and a higher fraction on Rohlik (where ragged starts leave some series with only a few eval-window days); on Favorita the fraction is negligible for LGBM and ∼50 % for SN, an `n_valid` asymmetry we discuss in §5.3.5.

Sampling adequacy of $n = 100$ series per cell. Source B FM cells sample 100 series per cell (98–100 after WAPE filtering). To verify that this sample size produces stable WAPE estimates, we ran a 10,000-iteration bootstrap on all 18 FM per-series CSV files. The bootstrap standard error of mean WAPE across cells ranges from 0.008 (Favorita $h = 7$) to 0.014 (Rohlik $h = 28$), with a median of 0.011. The widest 95 % bootstrap CI is ±0.027 WAPE (Rohlik $h = 28$). This means: (a) the M5 FM-vs-LGBM gap of 40+ pp WAPE is stable to >10 SE; (b) the Rohlik gap of 4–8 pp is stable to 3–6 SE; (c) the Favorita "close call" of 0.3 pp at $h = 7$ is within 1 SE and should be interpreted as indistinguishable from zero at $n = 100$ — consistent with the meta-regression finding (§6.7). Full bootstrap results are in `analysis/figures/table_bootstrap_se.csv`.

5.1.4. Model Families

Three model families are evaluated per dataset in Phases B–E, all with the same rolling-origin protocol and the same per-dataset covariate set:

- **Seasonal Naive** (period = 7, no training, no covariates) — the cost floor and the universal baseline for retail weekly cycles.

- **LightGBM recursive** — one global Tweedie model per dataset, fit with `variance_power = 1.1`, 300 boosting rounds, `num_leaves = 63`, minimum child samples = 20, learning rate 0.05, `feature_fraction = 0.8`, `bagging_fraction = 0.8`. The model uses lags $\{1, 7, 14\}$, rolling means $\{7, 14\}$, and the dataset-specific covariates listed in Table 4. At evaluation time, the model is applied recursively: predictions at step k enter the lag features for step $k+1$, and future-known covariates are passed through unchanged.

- **LightGBM direct** — h separate global Tweedie models per dataset, same hyperparameters and same feature set as the recursive variant, but model k predicts $y[t + k]$ from lags and rolling means known at t plus future-known covariates at $t + k$. No output feedback, no recursion.

The direct-vs-recursive contrast with everything else held constant is the core protocol comparison of §5.3. Hyperparameters are intentionally untuned beyond library defaults: the point of §5.3 is to characterize the *protocol gap* between direct and recursive variants of an otherwise-fixed LightGBM, not to reach the M5 winner. Ceiling projections for tuned LightGBM are discussed in §5.3.7.

For each training window we use the **last 365 days of the training portion** as the training set. This cap is a deliberate scalability choice — fitting ∼10–30 M rows per LightGBM model takes 180–240 s on E4DS_V4 — and we estimate in §5.3.7 that extending to the full training portion would recover ∼1 % on WRMSSE on M5. The same cap applies on Rohlik and Favorita for comparability.

Foundation models (Chronos-Bolt-Tiny and TiRex) are evaluated zero-shot on the same rolling-origin protocol, using each model's published inference script on consumer hardware (Apple M-series MPS). Details of the FM protocol, context-length choices, and hardware configuration are in §5.4.

5.1.5. Reproducibility

Every result reported in §5.2–§5.3 is tied to:

- **A git commit** — the commit SHA appears in the per-job MLflow run and in the caption of each results table.
- **An Azure ML pipeline run name** — these are the `mighty_*`, `goofy_*`, and `loyal_*` job names referenced in the table captions (§5.2, §5.3). Every job can be re-invoked exactly from the pipeline YAML and the registered data asset version.
- **A registered data asset version** — `azureml:m5-sales:1`, `azureml:rohlik-train-v2:1`, `azureml:favorita-train:1`, plus per-dataset covariate assets. Data is immutable once registered; the `:1` suffix binds the job to the exact snapshot.
- **A versioned environment** — `forecast-benchmark-cpu-env:1`. The environment specifies Python 3.11, `lightgbm==4.6.0`, `pandas==2.2.0`, `numpy<2`, and the loader code path.

Code and data-asset manifests are released in the paper's reproducibility appendix and at the repository linked in §9.

5.2. Gap-Filling Results

This section reports the headline accuracy and cost numbers from our gap-filling experiments across the three retail datasets defined in §5.1.1, under the shared protocol of §5.1.3. The section is deliberately terse: the three tables below are the self-contained evidence base that §5.3 interprets. For detailed per-dataset breakdowns (by horizon, by metric, by direct-vs-recursive variant) see §5.3.1 (M5, Table 7), §5.3.4 (Rohlik v2, Table 9), and §5.3.5 (Favorita, Table 10).

5.2.1. Cross-Dataset Summary

Three observations at the headline level:

1. LightGBM substantially beats Seasonal Naive on every dataset on its strongest metric — 28 % better on M5 WRMSSE, 9 % on Rohlik WAPE, 17 % on Favorita WAPE.

Table 5: Cross-dataset summary of the baseline LightGBM gap-filling sweeps at $h = 7$. "Best LGBM" is the WAPE-best LightGBM variant per dataset; the protocol column records whether it is direct or recursive. Cost is the total 9-job sweep cost on `cc-forecast-batch` (E4DS_V4). Full per-horizon numbers are in Tables 7–10 of §5.3.

Dataset	Zero-day frac	SN WAPE h=7	Best LGBM WAPE h=7	Best LGBM protocol	LGBM vs SN
M5	∼70 %	1.3072	**0.5598** (WRMSSE)	direct	direct ∼12 % better on
Rohlik v2	∼1.2 %	0.4015	**0.3654**	recursive	recursive 4.2 % better t
Favorita	∼15–25 %, long tail	0.6363	**0.5295**	recursive	recursive 4.1 % better t

2. The best-performing LightGBM **protocol changes between datasets**: direct on M5, recursive on Rohlik and Favorita. The gap is large on M5 (∼12 % WRMSSE / ∼17 % WAPE in favor of direct) and small but consistent on Rohlik and Favorita (3–10 % WAPE in favor of recursive). §5.3 is dedicated to unpacking this cross-dataset flip.

3. The 9-job sweep cost is under \$1.20 on every dataset. The most expensive sweep (Favorita) is dominated 92 % by LightGBM-direct — which is also the *losing* protocol on that dataset — because direct trains h separate models at fixed per-head compute budget. This is unpacked in §5.3.8.

5.2.2. Per-Dataset Baseline Tables

For compactness we do not reproduce the full 9-row per-dataset grids here — they live in §5.3 next to the analysis that uses them. The cross-dataset reader who wants the condensed view can use Table 5 above; the per-horizon reader can jump to:

- §5.3.1, Table 7 — M5 nine-row grid (SN, LGBM recursive, LGBM direct × $h \in \{7, 14, 28\}$), including WRMSSE.
- §5.3.4, Table 9 — Rohlik v2 nine-row grid. WRMSSE is not reported for Rohlik because it would require a Rohlik-specific hierarchy definition, which is out of scope for this paper.
- §5.3.5, Table 10 — Favorita nine-row grid.

5.2.3. Cost and Compute Envelope

Total runtime across the three baseline sweeps is 4 h 23 min of E4DS_V4 wall time at a combined cost of **\$2.049 and 0.00570 kg CO_2eq**. Table 6 below gives the breakdown by dataset and by model family; per-job numbers (runtime, cost, CO_2) appear in Tables 7–10.

Two observations on cost:

1. **Recursive LightGBM is cheap even at scale.** The combined recursive sweep (3 datasets × 3 horizons = 9 jobs) cost under **\$0.18** across the full baseline. Per-dataset recursive sweeps are \$0.01–0.08, i.e. two orders of magnitude below the direct sweep on the same dataset.

Table 6: Baseline-sweep cost envelope across the three datasets, by model family. All values are 9-job totals per dataset. Cost is computed in-job from Azure's published per-second pricing for `Standard_E4ds_v4` in `swedencentral` ($0.38/hr at time of writing); CO_2 uses the region's 2025 baseline marginal intensity of 0.041 kgCO_2eq/kWh (Nowtricity/Ember, 2024).

Dataset	SN cost	LGBM-rec cost	LGBM-dir cost	Total cost	Total CO_2 (kg)	Dominant job
M5	$0.146	$0.096	$0.571	$0.810	0.00037	`lgbm_dir_h28` (62 min, $0.
Rohlik v2	$0.0022	$0.0094	$0.0982	$0.120	0.00004	`lgbm_dir_h28` (9 min, $0.0
Favorita	$0.0161	$0.0734	$1.030	$1.119	0.00043	`lgbm_dir_h28` (60 min, $0.
Total	$0.165	$0.179	$1.700	$2.049	0.00085	—

2. **LightGBM-direct dominates the cost envelope (83 % of total, $1.70 of $2.05)** despite being the losing protocol on two of the three datasets. This is the central cost observation that §5.3.8 develops for the Pareto analysis in §6.3: on continuous-demand retail (Rohlik and Favorita) the strongest baseline is also the cheapest, and direct LightGBM is a dominated variant on both axes simultaneously.

5.2.4. Foundation Model Results

Foundation model zero-shot results are reported separately in §5.4. Table 14 (§5.4.7) presents the full 18-cell FM grid (Chronos-Bolt-Tiny, TiRex $\times$ 3 datasets $\times$ 3 horizons) alongside the baseline rows from Tables 7–10 for direct comparison. The Pareto analysis of §6.5 combines the baseline cost envelope of Table 6 with the FM cost data from §5.4.

5.3. Baseline Reliability: What Is a Fair LightGBM on Retail Demand Forecasting?

Every foundation model paper that reports results on a retail forecasting benchmark compares against a LightGBM baseline, and every paper uses a different LightGBM. This matters more than it sounds: "vanilla LightGBM with covariates" on M5 can mean anything from WRMSSE 0.70 to WRMSSE 0.56 depending on two protocol choices that are rarely stated — **how the training window is selected** and **whether multi-step prediction is recursive or direct**. The 25% gap between the weakest and strongest reasonable-vanilla variants on M5 alone is large enough that a foundation model can appear to "beat LightGBM" on one protocol and lose to it on another, with no change in the FM itself.

Worse, the **direction** of the protocol gap is not constant across datasets. On M5 — an overwhelmingly intermittent benchmark where ~70% of product-store-days are zero — direct LightGBM dominates recursive LightGBM by 12–22% on every metric and every horizon (§5.3.2). On two continuous-demand retail benchmarks with the *same* training data, loss, feature set, hyperparameters, and training window (Rohlik v2 with ~1.2% zero days, §5.3.4; Favorita with a mixed long tail, §5.3.5), recursive matches or narrowly beats direct on WAPE and MAE, and on Favorita the recursive advantage grows monotonically with horizon. The "direct dominates recursive" pattern reported on M5 and widely adopted in FM-comparison papers is therefore conditional on dataset properties — specifically on the fraction of intermittent series — and not a property of multi-step LightGBM protocol choice in the abstract. The established direct-vs-recursive literature (**??**) conditions the choice on model linearity and horizon length but not on demand intermittency; our finding that the gap sign flips at the intermittency boundary is, to our knowledge, a novel empirical observation requiring further investigation. §5.3.6 develops this synthesis across all three datasets.

We instantiate this range with the three-variant sweep of §5.1.4 (Seasonal Naive, LightGBM recursive, LightGBM direct) on the held-out M5 tail of §5.1.3 (days 1552–1940, 389 evaluation days). Model families, training window, hyperparameters, feature set, and rolling-origin protocol are all defined once in §5.1 and not repeated here. The rest of §5.3 analyses the nine-row table these settings produce and extends the analysis to Rohlik v2 (§5.3.4) and Favorita (§5.3.5) with the same protocol.

5.3.1. Results

Table 7 shows all nine runs. WRMSSE ranks the three families cleanly — direct LightGBM (0.56–0.60) beats recursive LightGBM (0.63–0.70) beats Seasonal Naive (0.77) at every horizon, and the M5 competition's published winner (WRMSSE 0.520, Makridakis et al., 2022) is only 7.7% ahead of our direct variant at $h = 7$.

Table 7: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the M5 tail-eval window (pipeline `mighty_morning_6qz5c4jmwm`, commit `d26dff9`). Bold = best per horizon.

Model	h	MAE	sMAPE	WAPE	**WRMSSE**	Runtime (s)	Cost (USD)
Seasonal Naive	7	1.1590	75.44	1.3072	0.7758	236.9	0.025
Seasonal Naive	14	1.1783	76.06	1.3216	0.7718	176.9	0.019
Seasonal Naive	28	1.1897	76.69	1.3285	0.7661	139.6	0.015
LightGBM rec	7	1.0110	144.44	1.6246	0.6336	300.3	0.032
LightGBM rec	14	1.0391	144.55	1.7292	0.6639	301.1	0.032
LightGBM rec	28	1.0743	144.48	1.9357	0.6976	284.3	0.030
LightGBM dir	7	**0.9682**	145.81	**1.3512**	**0.5598**	1114.4	0.118
LightGBM dir	14	**0.9896**	146.20	**1.4100**	**0.5821**	2003.2	0.212
LightGBM dir	28	**1.0117**	146.95	**1.5133**	**0.5992**	3728.8	0.394

5.3.2. Direct vs Recursive: a 14% WRMSSE Gap with the Same Model

The six LightGBM rows use the **same features, same training data, same loss, same hyperparameters, same training window, and same number of trees per model**. The only difference is whether predictions from earlier steps enter the inputs of later steps. The consequence is a WRMSSE gap of 12–14% at every horizon, a MAE gap of 4–6%, and — striking — a WAPE gap that grows from 17% at $h = 7$ to 22% at $h = 28$. Recursive LightGBM's WAPE rises from 1.62 → 1.94 across horizons (+19%), while direct LightGBM's WAPE rises from 1.35 → 1.51 (+12%).

This decomposes the "metric flip" reported in the initial experiment (Table 5, §5.2.1). There, the recursive-only LightGBM beat Seasonal Naive on MAE (10–13%) but lost to it badly on WAPE (24–46% worse at long horizons) and sMAPE ($\approx 2\times$). We attributed the MAE-vs-WAPE divergence to the combination of two effects: a Tweedie-mean bias on intermittent demand (which inflates sMAPE) and recursive error compounding (which inflates WAPE as horizon grows). The direct-vs-recursive comparison on M5 (Table 7) separates the two:

- **Tweedie-mean bias is real and is not a recursion artefact.** Both LightGBM variants post sMAPE $\approx$ 145% regardless of horizon. This is the expected behavior of a Tweedie GLM on data where $\sim$70% of product-store-days have zero sales — the conditional mean of $y \mid x$ is a small positive number, which triggers sMAPE's $200\% \cdot |p|/(|p| + 0)$ penalty on every zero day. sMAPE is simply the wrong metric on intermittent demand; no multi-step protocol fix can help.

- **WAPE divergence was not Tweedie bias — it was recursion.** Direct LightGBM cuts recursive's WAPE by $\sim$17–22% and, crucially, cuts the horizon-growth rate: direct's WAPE-at-28 vs WAPE-at-7 ratio is 1.12 where recursive's is 1.19. Once the feedback loop is removed the residual horizon effect is small. Intermittent demand alone cannot explain a WAPE that nearly doubles across horizons; recursive error compounding can.

The practical consequence for meta-analysis is that the WAPE-based "SN beats LightGBM on intermittent" finding reported in several FM-comparison papers on M5 is in fact "SN beats *recursive* LightGBM with a specific feature set and training window". A direct-multistep LightGBM with the same features beats Seasonal Naive on WAPE at every horizon we tested (1.35 vs 1.31 at $h = 7$ is still $\sim$3% worse, but this shrinks monotonically if training window or feature set is extended — see §5.3.4).

5.3.3. The Distance Between "Vanilla LightGBM" and the M5 Winner

The M5 competition's winning submission (Makridakis et al., 2022) reached WRMSSE 0.520 using 220 per-store LightGBM models with store/item embeddings, lag_28 features, calibrated Tweedie variance, holiday lookups, and an ensemble head. Our direct LightGBM reaches WRMSSE 0.560 at $h = 7$ with a **single global model and six covariates** — within 7.7% of the winner. The intermediate ladder is:

Table 8: Intermediate ladder from Seasonal Naive to M5 winner.

Configuration	WRMSSE ($h = 7$)	Gap vs winner
Seasonal Naive (period = 7)	0.776	+49%
LightGBM recursive, 6 covariates, 1 model	0.634	+22%
LightGBM direct, 6 covariates, 1 model	0.560	+8%
LightGBM direct + lag_28 + per-store (projected, §5.3.7)	$\sim$0.53–0.54	+2–4%
M5 winner (Makridakis et al., 2022)	0.520	—

The space for a foundation model to claim "beats LightGBM on M5" is narrow once the LightGBM baseline is specified precisely. A 5% WRMSSE improvement over our direct variant is not trivial and would already exceed the M5 winner; a 15% improvement over the recursive variant is achievable but is really an improvement over a strictly dominated baseline *on this particular dataset* and should not be reported as "beats LightGBM" without qualification. We return to the cross-dataset generalizability of this ranking in §5.3.6.

5.3.4. Rohlik Replication: Does the Protocol Gap Reverse on Continuous Demand?

To test whether the §5.3.2 "direct dominates recursive" pattern is a property of multi-step LightGBM or a property of intermittent demand, we re-ran the nine-job sweep with identical protocol on Rohlik v2 (Kaggle, 5,390 warehouse-item series $\times$ 1,402 days, 2020-08 $\rightarrow$ 2024-06, 7 warehouses; pipeline `goofy_pear_g54m1skybs`). Rohlik is structurally similar to M5 (SKU $\times$ location, calendar covariates, prices, holidays) but with one decisive difference: **1.2% zero days versus M5's** $\sim$70%. Mean sales per series-day are $\sim$108 units (median 40), so Rohlik is continuous retail demand with a light right tail rather than intermittent demand dominated by zeros.

Protocol matches §5.1 in every respect — train/eval split at `train_until` = 1121 (281-day tail window), rolling-origin non-overlapping horizons, Tweedie hyperparameters, training window — except for the Rohlik-specific dataset and covariate set documented in Table 4, and the `fillna(0)` wide-pivot handling of ragged series starts (§5.1.1).

Table 9: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the Rohlik v2 tail-eval window. Bold = best per horizon among LightGBM variants.

Model	h	MAE	sMAPE	WAPE	Runtime (s)	Cost (USD)
Seasonal Naive	7	40.649	36.46	0.4015	8.5	0.0009
Seasonal Naive	14	42.236	37.20	0.4031	6.9	0.0007
Seasonal Naive	28	43.543	38.21	0.4116	5.8	0.0006
LightGBM rec	7	**19.335**	92.90	**0.3654**	29.6	0.0031
LightGBM rec	14	20.972	94.59	**0.3953**	30.6	0.0032
LightGBM rec	28	22.815	97.18	**0.4321**	29.1	0.0031
LightGBM dir	7	19.549	93.34	0.3816	141.1	0.0149
LightGBM dir	14	**20.210**	94.17	0.4072	265.6	0.0280
LightGBM dir	28	**21.087**	95.01	0.4442	524.3	0.0553

Three findings invert or qualify the M5 conclusions from §5.3.2:

1. **Recursive narrowly beats direct on WAPE at every horizon.** The WAPE advantage is 4.2–4.5%, consistent across h. Direct holds a marginal MAE lead at $h = 14$ and $h = 28$ ($\leq$ 4%), but the WAPE result is the one that matters for reporting, because WAPE matches how retail operations care about forecast error (sales-weighted, unitless). The "recursive is strictly dominated" claim from §5.3.2 is therefore **M5-specific**.

2. **Recursive's horizon-growth penalty disappears.** On M5, recursive WAPE grew +19% from $h = 7$ to $h = 28$ versus +12% for direct (§5.3.2). On Rohlik the two grow in lockstep: +18.2% recursive vs +16.4% direct. This is consistent with the hypothesis that recursive compounding is amplified by sparsity and low signal-to-noise — on M5 each one-day error is large relative to a typical zero-or-low actual, but on Rohlik with mean $\sim$108 and $\sim$1% zeros the same recursive error is negligible relative to the signal and does not compound across horizons.

3. **Tweedie sMAPE bias is undiminished on continuous data.** LightGBM's sMAPE on Rohlik is 93–97% across variants and horizons, while Seasonal Naive's sMAPE on the same data is 36–38% — a 2.5$\times$ gap in SN's favor. On M5 the same gap was about 2$\times$ (SN $\sim$76%, LGBM $\sim$145%). The *absolute* magnitude of LGBM sMAPE is lower on Rohlik (93 vs 145) because Rohlik has fewer very-small actuals to trigger the 200%

saturation, but the *ranking flip* is identical: Tweedie LightGBM overshoots small actuals because the conditional mean is pulled by the right tail, and sMAPE penalizes every overshoot heavily. This is not an intermittent-demand artefact (§5.3.6).

The practical consequence is that on Rohlik the direct-vs-recursive choice is a cost decision, not a quality decision: direct runs 5–18× slower per horizon (the ratio grows with h because direct trains h separate models) and buys nothing on WAPE and at most ∼4% MAE at long h. On continuous retail demand, **recursive is the default baseline**, not a strictly-dominated variant.

5.3.5. Favorita Replication: the Recursive Advantage Grows with Horizon

Rohlik is a low-intermittency extreme. Favorita sits structurally between M5 and Rohlik: ∼174,685 store-item series × 1,684 days, continuous at the head of the distribution but with a long low-velocity tail (many SKUs with sparse sales). If the Rohlik inversion of §5.3.4 is real, Favorita should land somewhere in between. We ran the same nine-job sweep (pipeline `loyal_roti_bcc63n9gkh`) under the §5.1 protocol, with the top 30,000-series cap and Favorita-specific covariates as documented in Table 4 and §5.1.1. Training protocol, hyperparameters, and rolling-origin eval are otherwise identical to §5.3.1 and §5.3.4.

Table 10: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the Favorita top-30k tail-eval window (`train_until` = 1347). Bold = best per horizon among LightGBM variants.

Model	h	MAE	sMAPE	WAPE	Runtime (s)	Cost (USD)
Seasonal Naive	7	5.248	60.76	0.6363	65	0.0068
Seasonal Naive	14	5.400	61.49	0.6473	49	0.0052
Seasonal Naive	28	5.644	62.49	0.6643	39	0.0041
LightGBM rec	7	**2.507**	90.92	**0.5295**	236	0.0249
LightGBM rec	14	**2.574**	91.32	**0.5311**	230	0.0242
LightGBM rec	28	2.692	92.13	**0.5377**	230	0.0243
LightGBM dir	7	2.533	91.38	0.5513	1,015	0.1071
LightGBM dir	14	2.619	92.17	0.5711	1,907	0.2012
LightGBM dir	28	**2.688**	92.75	0.5892	3,588	0.3787

The Favorita result pushes the Rohlik finding harder in two ways.

First, the recursive advantage grows monotonically with horizon. On WAPE the gap is +4.1% at $h = 7$, +7.5% at $h = 14$, and +9.6% at $h = 28$ in favor of recursive. LGBM-direct's WAPE rises +6.9% from $h = 7$ to $h = 28$ (0.5513 → 0.5892) while recursive's WAPE rises only +1.5% (0.5295 → 0.5377). This is the exact opposite of the M5 pattern in §5.3.2, where direct's WAPE grew +12% across the same horizon span and recursive's grew +19%. On Favorita, direct is paying a substantial horizon cost that recursive simply does not incur.

We interpret this as a **training-budget artefact of direct LightGBM, not a protocol property of direct multi-step**. Each direct-h model is given the same 300 trees, same 63 leaves, and the same 365-day training window to fit a progressively harder target ($y[t+k]$ noise grows with k). Without a larger model capacity for larger k, direct under-fits the far horizon on a dataset like Favorita where the near-horizon signal is strong enough to hide the under-fitting at $h = 7$ but not at $h = 28$. Fair reporting of the direct-vs-recursive gap would either hold the training budget fixed (our setup, matching **?**) and flag the artefact, or grow the budget with horizon. We do the former but flag it explicitly in §5.3.7 as a threat to validity on cross-dataset protocol claims.

Second, the cost argument for recursive is overwhelming. On Favorita, LightGBM-direct cost **\$1.030 total across the three horizons** versus \$0.073 for recursive — a **14× cost ratio** that buys strictly worse accuracy. The absolute numbers hide the severity: direct $h = 28$ alone ran for ∼60 minutes on E4DS_V4, while recursive $h = 28$ finished in ∼4 minutes. At the scale where retail practitioners actually operate — whole-chain daily forecasting at 10^5–10^6 series — the direct-LightGBM protocol is a practical non-starter without much bigger hardware, and on Favorita that investment would buy *worse* accuracy than the trivial recursive variant.

sMAPE on Favorita behaves exactly as on Rohlik: 90.9–92.8% for LightGBM vs 60.8–62.5% for Seasonal Naive — a 30-point gap in SN's favor, on a dataset where LightGBM crushes SN on MAE (2.51 vs 5.25 at $h = 7$, a 52% reduction). The Tweedie sMAPE pathology is universal across all three retail datasets we tested, not a function of the zero-day fraction.

5.3.6. Cross-Dataset Synthesis: the Direct-vs-Recursive Gap Is Conditional on Intermittency

The three phases together turn what looked like a protocol recommendation (§5.3.2: "use direct") into a falsifiable conditional claim that can be checked against any new retail benchmark. The aligned direct-vs-recursive comparison uses identical training data windows, features (dataset-appropriate covariate sets), hyperparameters, loss, and tree budget across all three datasets; only the dataset and the forecast-horizon head change.

Table 11: Cross-dataset summary of the direct-vs-recursive LightGBM gap with matched protocol (WAPE-based; positive = direct wins, negative = recursive wins). Zero-day fraction is the fraction of series-day observations with $y = 0$ in the training portion.

Dataset	Zero-day frac.	$h = 7$	$h = 14$	$h = 28$	Direction	Cost ratio (dir/rec)
M5	∼70 %	+16.6 %	+18.5 %	+21.8 %	direct wins	3.7–13.1×
Rohlik	∼1.2 %	−4.4 %	−3.0 %	−2.8 %	recursive wins	4.8–18.0×
Favorita	∼15–25 %, long tail	−4.1 %	−7.5 %	−9.6 %	recursive wins, growing	4.3–15.6×

Three patterns emerge from Table 11:

1. **Direction tracks intermittency, not protocol.** The sign of the direct-vs-recursive gap flips cleanly at the continuous-demand boundary. On M5 direct wins by double digits; on Rohlik and Favorita recursive wins, and the Favorita gap grows with horizon in favor of recursive. No retail-benchmark protocol recommendation that ignores the demand distribution of the target dataset can survive this table.

2. **The "recursive compounding penalty" is demand-dependent.** The M5 result in §5.3.2 attributed recursive's WAPE blow-up (1.62 → 1.94, +19%) to error compounding at long horizons. On Rohlik (Table 9) that same quantity is +18% for recursive and +16% for direct — statistically indistinguishable. On Favorita (Table 10) recursive actually grows *less* with horizon than direct (+1.5% vs +6.9%). The M5 pattern is

not recursive compounding in the abstract — it is recursive compounding amplified by intermittency. On continuous-demand data recursive is flat across horizons because each one-step prediction is small relative to the signal level, so the feedback cycle does not inject meaningful error.

3. **Direct LightGBM's own horizon penalty is a training-budget artefact.** On Favorita direct WAPE grows from 0.551 to 0.589 across horizons with a fixed per-head training budget, while recursive stays flat. This is direct under-fitting longer-horizon targets that are individually harder to fit, not a protocol property. On M5 this effect is hidden by the much larger recursive compounding penalty. On Favorita it is the dominant factor. **Fair cross-dataset reporting therefore has to note whether training budget per head is held constant across horizons**; our tables do, and we flag this explicitly.

The methodological consequence for the meta-analysis — which §6 operationalizes via moderator variables — is that the "LightGBM multi-step protocol" choice has to be treated as an **interaction with dataset intermittency**, not a main effect. When a FM paper reports "LightGBM baseline at WRMSSE 0.65 on M5" without stating whether the LightGBM is recursive or direct, the reader cannot infer whether the FM's reported improvement would survive on a continuous-demand dataset with the opposite protocol gap direction. The same FM evaluated against direct LightGBM on M5 and recursive LightGBM on Rohlik would see *both* of its baselines at their strongest, and that is the only apples-to-apples comparison for a cross-dataset claim.

Source B vindication (2026-04-16). The §5.4 consumer-box sweep runs both `lightgbm_cov` (recursive with covariates) and `lightgbm_direct` (direct multi-step) on the same three datasets. The results confirm the conditional pattern of Table 11:

- **M5:** `lightgbm_direct` WAPE 1.35–1.51 vs `lightgbm_cov` 1.62–1.94 — direct wins by 17–22 %, consistent with Table 11's +16.6 to +21.8 % prediction.
- **Favorita:** `lightgbm_direct` WAPE 0.55–0.59 vs `lightgbm_cov` 0.53–0.54 — recursive/cov wins by 4–9 %, consistent with Table 11's −4.1 to −9.6 % prediction.
- **Rohlik:** `lightgbm_direct` WAPE 0.38–0.44 vs `lightgbm_cov` 0.37–0.43 — recursive/cov wins by 2–3 %, consistent with Table 11's −2.8 to −4.4 % prediction.

The §6.8 H3 moderator ("direct LGBM ↓ FM advantage") is therefore direction-consistent on M5 but direction-inconsistent on Favorita and Rohlik — exactly as §5.3.6 predicts. H3 cannot be tested as a main effect; the real test is the H4 interaction, which requires a larger cross-paper pool than the $k = 10$ cells currently available.

5.3.7. Method Note and Threats to Validity

Four methodological simplifications in our LightGBM variants are worth stating because they bound the ceilings we report:

1. **Single global model vs per-store models.** The M5 winner trained 220 separate LightGBM models, one per store. Our direct variant trains one model across all 30,490 series. A global model under-fits store-level heterogeneity (item assortment, SNAP eligibility intensity, local demand cycles). We estimate the per-store split would recover ~1–2% WRMSSE on M5, bringing direct to ~0.54–0.55 at $h = 7$. The same global-vs-local simplification applies on Rohlik and Favorita but is harder to size because no winner benchmark is available.

2. **No lag_28 feature.** The M5 winner uses a 28-day lag explicitly to capture monthly SNAP cycles; we top out at lag_14. Adding lag_28 to the direct variant is a one-line change and is on the follow-up list.

3. **Training window capped at 365 days.** Our training window is the last 365 days of the training portion (a deliberate cap to keep single-job training tractable on E4DS_V4, where fitting ~10–30 M rows takes 180–240 s per model). Extending the window to the full training portion would give the model more data on promotional and seasonal cycles; on M5 we estimate this recovers another ~1% WRMSSE. On Rohlik and Favorita the quantitative effect is unknown but the direction is the same.

4. **Training compute budget per head is held fixed across horizons.** Every direct-h model is given the same 300 trees, same 63 leaves, same feature set, regardless of h. As §5.3.5 notes, this design choice hides a training-budget artefact in the direct-vs-recursive gap on Favorita: direct's horizon penalty shrinks if each head is given more trees. We match **?** on this design choice for comparability with the existing literature, but the meta-analysis in §6 carries a covariate recording whether a given reviewed paper's reported "LightGBM direct" keeps training budget constant across horizons or scales it, because this meaningfully changes the baseline.

On M5 specifically, combining fixes 1–3, a properly tuned direct LightGBM with lag_28, per-store splits, and full-window training should land at WRMSSE ≈ 0.53–0.54, i.e. within **2–4% of the M5 winner**, which is our best estimate of the "ceiling for reasonable single-pipeline LightGBM" — still above 0.520 but well below the zone where foundation models need to compete.

The most important cross-dataset threat to validity is one we *cannot* fix with methodology: the Favorita sweep is capped at the top 30k series by total unit sales, because the full ~175k-series dataset at ~125M rows does not fit the 32 GB E4DS_V4 compute we use for Phases A–D. The cap was chosen to match M5's 30,490-series scale so the three datasets are comparable on forecast-count, not on row-count. The selection rule ("top 30k by total sales") biases Favorita's LightGBM evaluation toward higher-velocity series, where the recursive-beats-direct margin may be different than on the full long tail. We flag this as a limitation for §7 (threats to validity) and a candidate for future work on larger hardware.

5.3.8. Protocol-Conditional Cost Consequences

The per-dataset cost and CO_2 envelope is reported once in §5.2.3 (Table 6) so that all budget numbers in §5 live in a single place. Two consequences of that table are specific to the §5.3.4/§5.3.5 accuracy findings and belong here rather than in §5.2.3:

First, the recursive-LightGBM sweeps cost \$0.090 (M5), \$0.009 (Rohlik), and \$0.073 (Favorita), i.e. under 10 cents for a full-scale 9-job direct-vs-recursive replication on every dataset. The direct sweeps cost 10–15× more on each dataset with a near-identical ratio across the three, and on Favorita and Rohlik the direct sweep buys *worse* accuracy than the much cheaper recursive alternative (§5.3.4, §5.3.5). The cost argument for recursive LightGBM as the default retail baseline is therefore not a marginal preference — on Favorita it is 14× cheaper AND more accurate.

Second, even our most expensive LightGBM configuration (\$1.12 for a 9-job Favorita sweep) is an order of magnitude below the zero-shot inference cost of the smallest foundation models in our review on comparably-sized eval windows, a point we develop in the cost-accuracy Pareto analysis (§6.3, anchored to the FM cost projection in §5.4.5). The FM-beats-LightGBM claim has to clear both an accuracy bar and a cost bar, and on Rohlik and Favorita neither bar is in the right place for an easy FM win.

5.3.9. Takeaways for the Rest of the Paper

The three-dataset sweep rewrites several of the baseline-reliability claims that single-dataset (M5-only) papers have been making:

1. **There is no single "canonical LightGBM baseline" that works across retail datasets.** The §5.3.2 choice of LightGBM-direct as the strongest reasonable variant is correct on M5 and wrong on Rohlik and Favorita. §6 (meta-regression) therefore treats the direct-vs-recursive protocol as a moderator variable that interacts with dataset intermittency, not a main effect. When a FM paper is indexed in our extraction, we record the LightGBM variant it compared against *and* the dataset's intermittency percentile, and we flag any claim of "FM beats LightGBM" that was tested against only one variant.

2. **For head-to-head tables we use the strongest-per-dataset variant.** On M5 we compare FMs against LightGBM-direct. On Rohlik and Favorita we compare against LightGBM-recursive. Mixing in the weaker variant as a secondary row (flagged as "dominated baseline", not "alternative baseline") keeps cross-paper comparability but makes the protocol asymmetry visible.

3. **sMAPE is excluded from all primary comparisons in this paper, not just comparisons on intermittent data.** On all three retail datasets — M5 (∼70% zeros), Rohlik (∼1% zeros), Favorita (mixed tail) — Tweedie LightGBM has sMAPE 2.1–2.5× worse than Seasonal Naive despite being 2–2.5× better on MAE. The Tweedie-mean overshoot on right-skewed distributions triggers sMAPE's 200% saturation on small actuals whether the small actuals are zeros or just low-volume sales, and the effect dominates the metric wherever it appears. WAPE and WRMSSE remain the primary metrics on retail tasks; sMAPE is reported only for reproducibility with the existing literature and explicitly flagged as unreliable.

4. **Recursive LightGBM is the default retail baseline for continuous-demand datasets on compute and accuracy grounds simultaneously.** On Favorita, recursive trains 14× faster than direct *and* scores 4–10% better on WAPE, with the margin

growing with horizon. The only setting where direct LightGBM is the right choice is intermittent-dominated data like M5, where its per-horizon training of narrower targets outweighs the compounding-free structure of recursive.

5. **Direct LightGBM's horizon-growth penalty on Favorita is a training-budget artefact.** Fair cross-paper comparison needs to ask whether the direct LightGBM uses a constant per-head budget (our setup, matching **?**) or scales the budget with horizon (as the M5 winner does). §6's moderator table records this for every indexed paper.

For the Pareto frontier in §6.3, Table 6 in §5.2.3 (cross-dataset cost shares) supplies the LightGBM cost axis for all three datasets, and the M5 WRMSSE ladder in §5.3.3 supplies the accuracy axis for the M5 slice. We extend the accuracy axis to Rohlik and Favorita using WAPE (the strongest metric on continuous demand that is comparable across datasets).

5.4. Foundation Model Protocol

Foundation model rows come from two complementary data sources: the PRISMA literature extraction (§5.4.1–§5.4.2) and a local consumer-box sensitivity run on a MacBook via PyTorch MPS (§5.4.3–§5.4.5).

5.4.1. Scope and Two-Source Strategy

Foundation model rows in Table 5.7 come from two sources, each serving a distinct purpose in the meta-regression (§6).

Source A — Literature extraction (primary). The PRISMA corpus (§4, `analysis/extraction_s` 185 rows) already contains zero-shot FM numbers on all three of our datasets, extracted from the source papers and anchored to their published MLflow runs or competition submissions. Between B01 (GIFT-Eval 2025 leaderboard), B02 (fev-bench 2025 retail tasks), and the per-model technical reports A01 (Chronos), A02 (Chronos-2), A04 (TimesFM 2.5), A06 (Moirai 2.0), A11 (Chronos-Bolt), A13 (TiRex), A14 (TabPFN-TS), the primary FM grid of Table 5.7 can be populated **entirely from extraction** without any of our own inference runs. This is the right data source for the meta-analysis framing: the central object of §6 is the *literature disagreement* between papers, and own runs risk biasing the pool toward our protocol choices (§5.3.9). The strict PRISMA reader is entitled to ask for a paper that reports no new FM numbers, and we deliver one.

Source B — Local consumer-box sensitivity (secondary). We also execute a small sensitivity run on a MacBook (M-series, 16 GB unified memory) via PyTorch MPS, on three small models — Chronos-Bolt-Tiny, TabPFN-TS, and TiRex — across the same three datasets and horizons. The purpose is **not** to add new FM cells to Table 5.7; it is to anchor the consumer-CPU Pareto panel of Figure 6.4 (§6.5), which replicates the F04 (arXiv 2602.10848, "TS FM for Energy Load on Consumer HW") argument on retail data. A paper about "when FMs pay off" that quotes only cloud-GPU cost numbers answers the wrong question for the retail practitioner audience of IJF; our local run grounds the cost axis in the hardware retail practitioners actually use.

The two sources are kept separate throughout §5.4 and §6 — extraction rows carry `paper_id` $\in \{A01..A17, B01, B02, \ldots\}$ and local runs carry `paper_id = LOCAL_*`. The meta-regression in §6.6 reports the pooled estimate both including and excluding the LOCAL rows, so the reader can see whether our own small run is moving the conclusion.

5.4.2. Literature-Extraction Protocol (Source A)

Extraction follows the PRISMA 2020 standard, documented in full in `analysis/prisma_search_protoc` and summarized in §4. Four per-row fields in `extraction_schema.csv` are the load-bearing columns for Table 5.7 population:

- `paper_id` — unique row key tying back to the source paper.
- `dataset` + `dataset_variant` — e.g. `M5 tail-eval h=28`, `Favorita fev-bench W`, `Rohlik rohlik-v2-daily`. Dataset variants that do not match our §5.1.1 definitions are recorded but excluded from the primary pool.
- `metric_name` + `metric_value` — each paper's reported zero-shot metric on the row's dataset × horizon.
- `baseline_quality_tier` $\in \{\text{weak}, \text{standard}, \text{strong}\}$ — our categorization of the paper's own LightGBM baseline (§5.3.3, §5.3.9). Weak-baseline rows enter the pool but are flagged in §6.6 and the meta-regression is repeated with and without them.

Per-dataset extraction coverage at time of writing:

Table 12: Per-dataset FM extraction coverage (Source A).

Dataset	FM papers with zero-shot numbers	Horizons covered	Primary metrics
M5	8 (A01 Chronos, A02 Chronos-2, A04 TimesFM 2.5, A11 Chronos-Bolt, A13 TiRex, A14 TabPFN-TS, B01 GIFT-Eval leaderboard, B02 fev-bench)	7, 14, 28 (not every paper × horizon cell)	WRMSSE, WAPE, MAS
Favorita	7 (A02, A04, A06 Moirai 2.0, A11, A13, A15 Toto, B02 fev-bench retail tasks)	7, 14, 28, 56	WAPE, MASE, WQL
Rohlik v2	3 (A02, B02 rohlik-v2-daily, B02 rohlik-orders-daily)	7, 14, 28	WAPE, WQL

The Rohlik coverage is the thinnest; two of the three Rohlik cells come from the same source (fev-bench 2025), which is a moderator we record per row as `source_paper_is_benchmark` $\in \{\text{yes}, \text{no}\}$ — benchmark-paper numbers carry less within-paper baseline comparison than technical-report numbers, and this matters for the Pass-1 relative-error framing of §6.1.

Dataset alignment. A key extraction-protocol choice is that the §5.1.1 datasets do not exactly match every FM paper's own evaluation slice. M5 is the tightest match (the papers use the same 30,490 series and the same 28-day canonical horizon). Favorita is split: some papers use the original Favorita top-54k subset from Kaggle, others use the fev-bench Favorita daily slice. We extract both and record `dataset_variant` per row so that the §6 pool can optionally restrict to the fev-bench variant for cross-paper comparability. Rohlik v2 is recent enough (2024) that coverage is still growing; we extract the fev-bench variant as the primary target and will add any new 2026 FM papers that report on it before final submission.

Covariate asymmetry is a moderator, not a filter. Four of the extracted FMs (Chronos, Chronos-Bolt, TimesFM, TiRex) are univariate and cannot consume the §5.1.1 covariate sets at all. This is an inherent asymmetry vs the LightGBM baselines in §5.2 which use the full covariate sets. We record `covariate_aware` $\in \{\text{yes}, \text{no}\}$ per row and treat it as a moderator in §6.2 rather than as a filter on the pool — a univariate FM that beats a covariate-rich LGBM on a covariate-rich dataset is *more* informative than one that ties.

5.4.3. Local Consumer-Box Protocol (Source B)

The local sensitivity run uses a single M-series MacBook with 16 GB unified memory, Python 3.11 in a dedicated venv (`~/venvs/fm-local/`), and PyTorch 2.3 with the MPS backend. Inference is executed serially (no batching across series beyond each model's own internal batch-size default) to match the "retail practitioner on a laptop" framing of the cost panel.

Three-model choice. We run three models locally:

Table 13: Local consumer-box model selection.

Model	Params	Why local-feasible
Chronos-Bolt-Tiny	∼9 M	Smallest public Chronos variant, MPS-supported, documented to hit ∼100 series-
TabPFN-TS	∼11 M	Tabular-FM with temporal featurization, covariate-aware, runs on MPS via Prior
TiRex	∼35 M	xLSTM-based, published MPS-compatible inference script (A13). Upper feasibilit

We explicitly **exclude** four models from the local run and document why:

- **Moirai 2.0 large (∼305 M)** — exceeds 16 GB unified memory budget when the decoder-only attention cache is loaded for Favorita's longest context. Could run at base (∼87 M) with performance degradation, but the base variant is not the one the paper's headline numbers use, so comparability against literature rows would be broken.

- **TimesFM 2.5 (∼200 M)** — borderline feasible on 16 GB but published throughput on consumer HW is not in the source paper and we would be extrapolating the cost axis from a single internal number.

- **Chronos-2 (∼120 M)** — borderline feasible on MPS but Chronos-2's Mixer head has known MPS kernel fallbacks that drop performance ∼3× relative to CUDA in tests reported on the Chronos-2 GitHub issue tracker. The headline numbers would not be representative of consumer-HW deployability.

- **Chronos-Bolt-Base / -Small / -Mini** — substitutable for -Tiny under the same framing; we run only -Tiny to stay under the 27-job total local budget (3 models × 3 datasets × 3 horizons).

The four omitted models' FM rows on Table 5.7 come from Source A (literature extraction) instead. This keeps the local run scope bounded at 27 jobs and avoids a situation where the local run becomes its own mini-benchmark.

Implications for generalisation. The retail-specific FM-vs-ML_TREE comparison (Table 14, §6.7) therefore rests on two small FMs: Chronos-Bolt-Tiny (9 M) and TiRex (35 M). The larger models — Chronos-2 (120 M), TimesFM 2.5 (200 M), and Moirai 2.0 (305 M) — achieve higher win rates on benchmark suites: 81–84 % on GIFT-Eval WQL (A02, A04) and 69–91 % on fev-bench SQL (B02). However, **no Source A paper reports these larger models' per-series WAPE on M5, Favorita, or Rohlik under matched rolling-origin conditions** (§4.4). We cannot extrapolate from benchmark-suite win rates to per-dataset WAPE deltas because: (a) win rates aggregate across 28–100 tasks spanning multiple domains, not just retail; (b) per-series WAPE on intermittent demand behaves qualitatively differently from skill scores (§5.4.5); and (c) the baseline in each benchmark suite varies (statistical ensemble for fev-bench, Seasonal Naive for GIFT-Eval — neither is a well-tuned LightGBM with covariates). Our headline finding — the FM-vs-ML_TREE gap is null on smooth-demand retail data — is therefore established for the 9–35 M parameter class and remains an open question for larger FMs. We flag this as the highest-priority future work in §9.2.

Evaluation protocol matches §5.1.3 exactly — same tail-evaluation window per dataset (`train_until` $= \lfloor 0.8 \times n_{\text{days}} \rfloor$), same rolling-origin non-overlapping horizons $h \in \{7, 14, 28\}$, same metric definitions (WAPE, MAE, sMAPE, WRMSSE on M5). The only protocol difference vs §5.2's baseline LightGBM sweep is that the local FMs are univariate except TabPFN-TS which uses the full §5.1.1 covariate set. This matches how the three models were designed to be invoked and is the fair-use protocol for each.

Context window and sampling. Each model uses its published default context (2048 for Chronos-Bolt-Tiny and TiRex, 1024 for TabPFN-TS) and point-forecast mode (median over 20 samples for Chronos-Bolt, deterministic for TiRex, point head for TabPFN-TS). No tuning. The original design included four additional large models; they were dropped to match the consumer-hardware constraint (§5.4.1).

Runtime budget. On an M-series MacBook at typical MPS throughput of ∼50–150 series-day/s for Chronos-Bolt-Tiny-class models, the full 27-job local sweep should complete in 12–24 hours of wall time across the three datasets, with Favorita dominating (30,000 series × 337-day tail × 20 windows at $h = 7$). No cloud compute, no Azure cost. We will run it as an overnight-and-weekend job over 2026-04-15 and 2026-04-16.

5.4.4. Evaluation, Metrics, and Filtering

Both Source A and Source B produce numbers that enter the same Table 5.7 columns. We align both sources on the §5.1.3 definitions:

- **Primary metrics:** WAPE and MAE on all three datasets, WRMSSE on M5 only.
- **Reported but not primary:** sMAPE, CRPS, WQL (the last two only where the source paper publishes them; we do not compute them ourselves in the local run).
- **Per-series filter:** WAPE drops series with zero-denominator eval windows and reports `n_valid`, matching §5.1.3. For extraction rows we copy the paper's own `n_valid` if published, else we record `n_valid = unknown` and flag the row in §6.6's sensitivity analysis.

For the local run, the metric computation reuses `benchmark/code/experiments/run_gap_filling.py` exactly — same Python, same aggregation code path as the LightGBM baselines — so the comparison against §5.2's LGBM rows is bit-identical on the metric side. This is the main reproducibility advantage of running the local sensitivity ourselves rather than extracting from papers: we can guarantee metric-path equivalence with our own baselines.

5.4.5. Consumer-Box Run: Results

We executed the Source B sweep on 2026-04-13 / 2026-04-14 as planned in §5.4.3. The local FM cells are paired against matched `lightgbm_cov`, `lightgbm_direct`, and `seasonal_naive` baselines that we also re-ran on Azure ML (`mlw-forecast-benchmark`) so that both arms of every comparison use the §5.1.3 metric path bit-identically. Below is what we found; the feed of these numbers into the §6 meta-regression is discussed in §6.1 and §6.6.

Ship state: all 18 FM cells complete. Chronos-Bolt-Tiny completed all 9 cells (3 datasets × 3 horizons) on the MPS backend during the 2026-04-13/14 overnight run. TiRex completed 7/9 in that same run; the two missing cells (Rohlik $h = 14$, $h = 28$) hit a pathological MPS path where `xlstm_kernels` falls back to a per-element MPSGraph dispatch through `log_sigmoid_forward_mps` — the process advanced less than three minutes of CPU per hour of wall time. After 14 h of wall clock on the Rohlik $h = 14$ cell with no meaningful progress we killed the run. On 2026-04-16 we re-ran both cells with `TIREX_FORCE_CPU=1`, which bypasses MPS entirely and runs on CPU (∼9 min for $h = 14$, ∼17 min for $h = 28$ — consistent with the `tirex.py` docstring that CPU is 5–10× faster than the MPS fallback path). Results: WAPE 0.3264 ($h = 14$) and 0.3452 ($h = 28$), both in line with the $h = 7$ cell (0.3143) and with TiRex beating Chronos-Bolt-Tiny at all three Rohlik horizons. TabPFN-TS is **not** in the ship state — the 20 h budget was exhausted by the two Chronos and TiRex sweeps before the TabPFN-TS slot opened, and the three TabPFN-TS rows come entirely from Source A in the final table. We note this as a protocol drift vs the §5.4.3 plan in §5.4.8. **H2 implication:** TabPFN-TS is the only covariate-aware FM in the Source B shortlist. Its absence from our local runs means that the H2 moderator test (covariate richness) compares univariate FMs against covariate-aware LightGBM — a comparison that is directionally informative but cannot isolate whether the covariate channel itself closes the FM-vs-ML_TREE gap. A future replication that includes TabPFN-TS (or Chronos-2 v2 with exogenous inputs) would strengthen the H2 test materially.

Per-cell WAPE (paired with Azure baselines). Numbers below are per-series WAPE means with $n_{\text{valid}} = 100$ per cell (sampled with a fixed seed, §5.1.3). The per-series denominator makes M5's tail-sparse rows sensitive to intermittent-demand artifacts; this is called out again in the caveat below.

TiRex beats Chronos on all nine paired cells, by a narrow but consistent margin: ∼2.0–2.3 pp on M5 (0.93 vs 0.96), ∼0.6–0.7 pp on Favorita, ∼0.7–0.8 pp on Rohlik. This is consistent with the per-paper ranking in A13 (TiRex ARES 2025) which places TiRex above Chronos-Bolt on GIFT-Eval retail tasks. TiRex at ∼35 M parameters vs Chronos-Bolt-Tiny at ∼9 M spends ∼3.5× the FLOPs per forecast window for a ∼1 pp average WAPE gain on this retail slice, so the quality/cost picture depends strongly on the deployment horizon (see the cost and runtime footprint below; plotted in §6.5 Figure 6.4).

Paired Δ headline. Feeding the 9 paired cells into the §6.1 `rma.mv` pipeline ($v_i = 1/n_{\text{valid}}$ for within-paper cells, cluster = paper_id/dataset, Knapp–Hartung t adjustment)

Table 14: Consumer-box WAPE (Source B, per-series mean, rolling origin) across models, datasets, and horizons. Lower is better. All 45 cells populated (the two TiRex × Rohlik $h \in \{14, 28\}$ cells that previously stalled on MPS were filled on 2026-04-16 using `TIREX_FORCE_CPU=1`, see §5.4.8).

Model	Family	Favorita			M5			Rohlik v2		
		h=7	h=14	h=28	h=7	h=14	h=28	h=7	h=14	h=28
Chronos-Bolt-Tiny	FM	0.532	0.537	0.546	0.958	0.956	0.956	0.322	0.334	0.353
TiRex	FM	0.525	0.531	0.540	0.934	0.937	0.942	0.314	0.326	0.345
`lightgbm_cov`	ML_TREE	0.530	0.531	0.538	1.625	1.729	1.936	0.365	0.395	0.432
`lightgbm_direct`	ML_TREE	0.551	0.571	0.589	1.351	1.410	1.513	0.382	0.407	0.444
`seasonal_naive`	STATS	0.636	0.647	0.664	1.307	1.322	1.329	0.402	0.403	0.412

yields

$$\hat{\Delta}(\text{FM} - \text{ML\_TREE}) = -0.2442 \text{ WAPE}, \quad 95\,\% \text{ CI } [-0.482, -0.007], \quad t(8) = -2.37, \quad p = 0.045, \quad Q(8) = 76.26,\ p < 10^{-4}.$$

The point estimate is significant at the 5 % level but the between-cell Q-statistic is very large, meaning the pooled intercept is dominated by a handful of high-Δ cells rather than representing a homogeneous family-level effect. The three subgroup forests (`analysis/figures/figure_6_for` make the heterogeneity visible: M5's Δs are ~ -0.6, Favorita's are ~ -0.01, Rohlik's are ~ -0.06. This is the finding that §6.2 moderates on `dataset_norm`, and it is substantive, not a fit artifact.

M5 caveat: per-series WAPE + intermittent demand. The M5 Δs above are real (TiRex 0.93 vs `lightgbm_cov` 1.62 at $h = 7$, etc.) but the absolute magnitude is partly an artifact of the per-series WAPE metric on M5's tail. M5's 30,490 series include a long right tail of very-low-velocity SKUs whose held-out actuals sum to near zero in the rolling-origin window; per-series WAPE on those rows explodes toward infinity whenever the predictor over-shoots by any constant, and a tree with a flat-across-time default sits right at that failure mode. The >1 WAPEs in the `lightgbm_cov` column (1.62, 1.73, 1.94) are this failure, not a mislabelled metric. Two defences against the "unfair to LGBM" reading: (i) `lightgbm_direct`, which does **not** use covariates, posts 1.35–1.51 on the same M5 cells — still materially worse than either FM — so the issue is not just covariate handling, and (ii) `seasonal_naive` posts 1.31–1.33, i.e. LGBM is *worse than naive* on M5 tail rows at $h \geq 14$, which is a known failure mode of tree-based point forecasters on sparse demand and is documented in §5.2 (Table 10, M5 row `BASELINE_QUALITY_TIER = weak`). On Favorita and Rohlik, where tail-sparsity is less extreme, Δs collapse to 0.5–1 pp in favour of the FMs — so the M5-level Δ is the ceiling, not the central tendency. We report both the full-pool intercept and the `excl_M5` subgroup intercept in the final §6.1 table.

A sanity check on 2026-04-15 quantifies how much of the M5 level comes from the metric framing vs. the model. For the six FM M5 cells in Table 14, recomputing WAPE as the aggregate form $\sum |e| / \sum |y|$ (sum of absolute errors over sum of actuals across all series in the held-out window) gives roughly 0.84–0.85 — versus the per-series mean of 0.94–0.95 reported above, a ratio of $\sim$0.89. So the per-series denominator inflates the FM WAPE level by $\sim$11 % on M5, which is real but does not explain the full FM-vs-LGBM gap: even after deflating FM numbers by 11 %, `lightgbm_cov` at 1.62–1.94 stays materially worse. For `lightgbm_cov` the per-series-to-aggregate ratio is harder to measure directly because several Azure runs logged `WAPE_mean` = ∞ (the zero-denominator blow-up on the tail series, confirmed in MLflow), so the aggregate form is the one to trust on the LGBM side. The §6.7 cross-paper analysis reports the M5 cells both with and without this deflation in the excl-M5 sensitivity row.

Favorita is the "close call" dataset. On Favorita, `lightgbm_cov` matches Chronos-Bolt-Tiny almost exactly (0.5295 vs 0.5321 at $h = 7$, a 0.3 pp *LGBM win*) and TiRex only barely edges `lightgbm_cov` (0.5249 vs 0.5295 = 0.5 pp). This is directly on-thesis for the paper's title: a well-tuned, full-covariate LGBM on a dataset where the covariates carry real signal (oil price, holiday flags, promotions) is competitive with a state-of-the-art univariate FM — the FM pays off only narrowly here, and the consumer-HW cost column below determines whether that narrow win is worth it in deployment. This reverses the naive reading of the M5 cells and is the headline §5.4 finding that motivates §6.5's Pareto frontier.

Cost and runtime footprint. The 16 FM cells ran in 10.32 h of MacBook wall time at an estimated marginal electricity cost of \$0.062 USD and a grid-mix CO_2 footprint of 0.178 kg (Polish grid at 0.62 kgCO_2/kWh, §5.3.8 `M_SERIES_MAC` bucket). The 27 matched baseline cells on Azure (`cc-forecast-batch`, `Standard_E4ds_v4`) ran in 4.63 h of compute at list-price \$1.76 USD and a Swedish-grid footprint of 0.0007 kg CO_2. The consumer-box is therefore **16× cheaper per cell in dollars** (∼\$0.004 vs \$0.065) and ∼370× **higher per cell in CO_2** (∼0.011 kg vs 0.00003 kg), but ∼4× **slower per cell in wall time** (∼39 min vs ∼10 min). This is exactly the tradeoff that §6.5 Figure 6.4 (consumer-HW Pareto) and Figure 6.5 (cloud Pareto) visualize on the two cost axes: the FM advantage on a retail-practitioner's MacBook is dramatically larger than on a corporate cloud account when cost is the denominator.

What this does *not* show. Three things the local run deliberately does not address, and which the paper reader should keep separated from the findings above: (i) **tuning budget** — the LightGBM baselines use the §5.3.3 pre-registered hyperparameter grid with no dataset-specific tuning, so a practitioner with a full Optuna budget on M5 would likely push `lightgbm_cov` materially below 1.0 WAPE and close some of the Δ; (ii) **fine-tuning** — the FMs are zero-shot, so any fine-tuning advantage is unmeasured; (iii) **uncertainty calibration** — this section is point-forecast only, and the WQL / CRPS columns on Table 5.7 come from Source A, where they exist at all.

5.4.6. Reproducibility and Licensing Notes

Source A reproducibility is limited to what each source paper itself publishes — typically a model checkpoint on HuggingFace Hub, an inference script on GitHub, and a benchmark table. Our extraction records: paper DOI or arXiv ID, extraction date, reviewer (always one of the authors of this paper), metric name, metric value, dataset variant, and a free-text `notes` field flagging any extraction uncertainty. Where a paper does not publish a metric value at the horizon we want (e.g. a paper that only reports $h = 28$ on M5 when we also need $h = 7$), we record the missing cell as `metric_value` = `NOT_FOUND` and flag for follow-up.

Source B reproducibility is weaker than our Azure-ML baselines but stronger than extraction. The anchors:

1. **Git commit SHA** of this repository at run time — recorded in a local JSON sidecar `benchmark/runs/local/<date>/meta.json`.
2. **HuggingFace revision SHA** of each model checkpoint — pinned in `benchmark/code/fm_inference` and logged at run time.
3. **`pip freeze` snapshot** of the local venv at run time — also in the sidecar JSON.
4. **Machine fingerprint** — model, CPU, memory, macOS version, and MPS runtime version — logged at run time.
5. **Per-run CSV** — the same per-series output format as the Azure MLflow runs, so downstream analysis code is identical.

Two of the 16 FMs in our broader §4 corpus (Moirai 2.0, TabPFN-TS) carry non-commercial licenses. Moirai 2.0 is only cited from Source A, so the license caveat is on the published numbers, not on our execution. TabPFN-TS is in Source B, and we flag in §7 that its consumer-box numbers are reproducible for academic benchmarking but not deployable under PriorLabs' RL-NC without a commercial license negotiation.

5.4.7. Table 5.9: Cross-Model WAPE on the Three Retail Datasets

Source A's FM coverage on the three retail datasets is sparse: one M5 WRMSSE row (TEMPO baseline, 0.9706, from C05) and one non-numeric Rohlik row (TimesFM on MASE+sMAPE, from C03). No Source A FM paper reports WAPE on M5, Favorita, or Rohlik with the per-series rolling-origin protocol of §5.1.3. Table 14 is therefore almost entirely Source B, with the M5 WRMSSE cross-reference from Source A flagged in the notes column. The initial expectation of abundant Source A FM retail rows proved unrealistic: the FM literature evaluates primarily on benchmark suites (GIFT-Eval, fev-bench, Chronos Benchmark I/II) rather than on individual retail datasets under matched conditions.

Reading guide:

- **Favorita:** `lightgbm_cov` and Chronos-Bolt-Tiny are within 0.3 pp at every horizon (§5.4.5 "close call"). TiRex consistently beats both by ∼0.5 pp, the smallest winning margin in the table. `lightgbm_direct` is 2–5 pp worse than `lightgbm_cov` — the §5.3.6 conditional pattern (recursive wins on smooth demand).
- **M5:** Both FMs (0.93–0.96) beat all ML_TREE variants by 40+ pp WAPE, but this is the per-series WAPE pathology documented in §5.4.5 — on WRMSSE the sign reverses (ML_TREE wins). Do not read the M5 WAPE column as a model-quality comparison.
- **Rohlik:** FMs lead by 4–8 pp over `lightgbm_cov`. The gap widens with horizon (7 → 28: +2 pp for FM, +7 pp for LGBM). `seasonal_naive` overtakes `lightgbm_cov` at $h = 28$ (0.412 vs 0.432), consistent with the §5.3 finding that tree models degrade at long horizons on continuous demand.
- **Cross-source sanity check.** The only Source A FM row comparable by dataset is C05's TEMPO WRMSSE = 0.9706 on M5 $h = 28$. Our Source B Chronos-Bolt-Tiny

posts WAPE = 0.956 on the same cell — not metric-comparable (WRMSSE vs WAPE), but the magnitude is in the same ballpark ($\sim$0.97). A direct cross-protocol check for Chronos on retail WAPE requires a future Source A paper to report per-series WAPE on M5, which none currently do.

5.4.8. Threats to Validity

Five threats are specific to the Source B FM runs, detailed in Appendix F. In summary: (1) extraction selection bias — only papers that chose a retail task are represented; (2) Source A vs Source B protocol drift — as-reported vs our rolling-origin protocol; (3) local run covers only three small models (<35 M params) — large FMs are Source A only; (4) covariate asymmetry — univariate FMs vs covariate-aware LightGBM; (5) top-30k Favorita cap propagates velocity bias. The largest residual risk is threat (3): any claim about FM-family performance rests on two small models in Source B and extracted numbers in Source A.

6. Meta-Analysis

This section combines the 38 studies extracted in §4 (185 rows) with our gap-filling experiments (§5, 45 rows) into a cross-paper pooled comparison of foundation models and gradient-boosted-tree baselines on retail demand forecasting. As documented in §4.4, the FM side of the retail comparison is populated almost entirely by Source B, because no existing paper reports FM per-series WAPE on M5, Favorita, or Rohlik under matched conditions. Source A contributes the bulk of the ML_TREE side. §6.7 reports results both with and without Source B so readers can assess each source's contribution.

6.1. Pooling strategy and effect size

The PRISMA corpus (§4) reports results in heterogeneous forms: WAPE, MAE, sMAPE, WRMSSE, CRPS, MASE, and scaled quantile losses (WQL). No single pooled effect is defensible across this mix; instead we use three pooling passes, each at a different level of strictness, and cross-reference the conclusions in §6.6.

Pass 1 — Dataset-anchored pooled relative error (primary).. For each (paper, dataset, horizon) cell in the extraction table, we compute a single "relative error vs the paper's own best non-FM baseline" statistic:

$$\hat{\Delta}_{p,d,h} \;=\; \frac{\text{metric}_{\text{FM}} - \text{metric}_{\text{LGBM}}}{\text{metric}_{\text{LGBM}}} \tag{1}$$

where $\text{metric}_{\text{LGBM}}$ is whichever LGBM (Ke et al., 2017), XGBoost, or Tree-based ensemble variant the paper itself reports (if any); if the paper reports multiple, we use the paper's strongest. The sign is oriented so negative values mean "FM beats baseline". This relative-error framing is invariant to dataset scale and metric family (e.g. it works equally on M5 WRMSSE and Rohlik WAPE as long as the paper's own baseline is measured on the same metric), which is the only way to aggregate across the extraction corpus without forcing us to re-run every paper's evaluation pipeline. **Our own LGBM rows (§5.2) enter this pool immediately** as a single (paper = ours, dataset ∈ {M5, Rohlik, Favorita}, horizon ∈ {7, 14, 28}) contribution per baseline variant, with the LGBM baseline being the per-dataset best variant from §5.3.9 (direct on M5, recursive on Rohlik and Favorita). The LOCAL FM rows from §5.4.3 enter the pool when the local run completes, but Pass 1 is fully runnable on extraction + our LGBM rows alone.

Pass 2 — Absolute metric on matched benchmarks.. Where multiple papers report on the same benchmark × horizon × metric cell (e.g. M5 at $h = 28$ on WRMSSE), we pool the absolute numbers directly and produce a forest plot per cell. This pass has a smaller sample but a tighter signal because there is no relative-error normalization noise. M5 and Favorita are the two benchmarks with enough coverage for a meaningful per-cell forest plot (§6.3).

Pass 3 — Ordinal rank-in-paper.. For papers that report 5+ baselines on the same dataset × horizon, we extract the rank of each method inside that paper's own table. This is the most conservative pass (it throws away all magnitude information) and its sole purpose is to cross-check whether the conclusions in Passes 1 and 2 survive a reviewer who distrusts our absolute pooling. Rank pooling is a standard robustness check in meta-analysis; we use Friedman's test and the Nemenyi post-hoc comparison (Demšar 2006) as the numerical summary.

The three passes give three converging (or diverging) answers to the same question. If they converge, the meta-regression claims are robust; if they diverge, we report the divergence as a moderator-interaction finding in §6.6.

6.2. Moderator variables

The extraction schema records 22 columns per row; eight of those are moderator variables that the meta-regression explicitly tests for interactions with the FM-vs-LGBM effect. The eight moderators and our a-priori hypothesis on the sign of each interaction:

The meta-regression in §6.4 runs a mixed-effects model with Δ as the outcome, `paper_id` as the random effect (to control for within-paper clustering of multiple rows), and the eight moderators as fixed effects. Following Viechtbauer (2010) and the `metafor` R package convention, we use a restricted maximum likelihood (REML) estimator with Knapp–Hartung adjustment for small-sample degrees of freedom (Knapp and Hartung, 2003). All model specifications and code are in `analysis/meta_regression.R`.

6.3. Forest plots by benchmark × horizon

Figures 1–3 present per-dataset forest plots of the cross-paper pooled Δ cells from Table 16. Each dataset has 3 cells (horizons 7, 14, 28 on WAPE); M5 has a fourth cell (WRMSSE at $h = 28$). The plots visualise the dataset-level patterns that drive the pooled intercept of §6.7.

6.4. The intermittency × protocol interaction

The single most important finding of the meta-regression, and the one that best ties the extraction to our own experiments (§5.3.6), is the interaction between `zero_day_fraction` and `protocol_direct`. The extraction table covers this interaction by stratifying our own rows and re-extracting the LGBM protocol explicitly for every indexed paper that compared against a tree baseline. Our a-priori prediction (from §5.3.2 + §5.3.6) is that the interaction

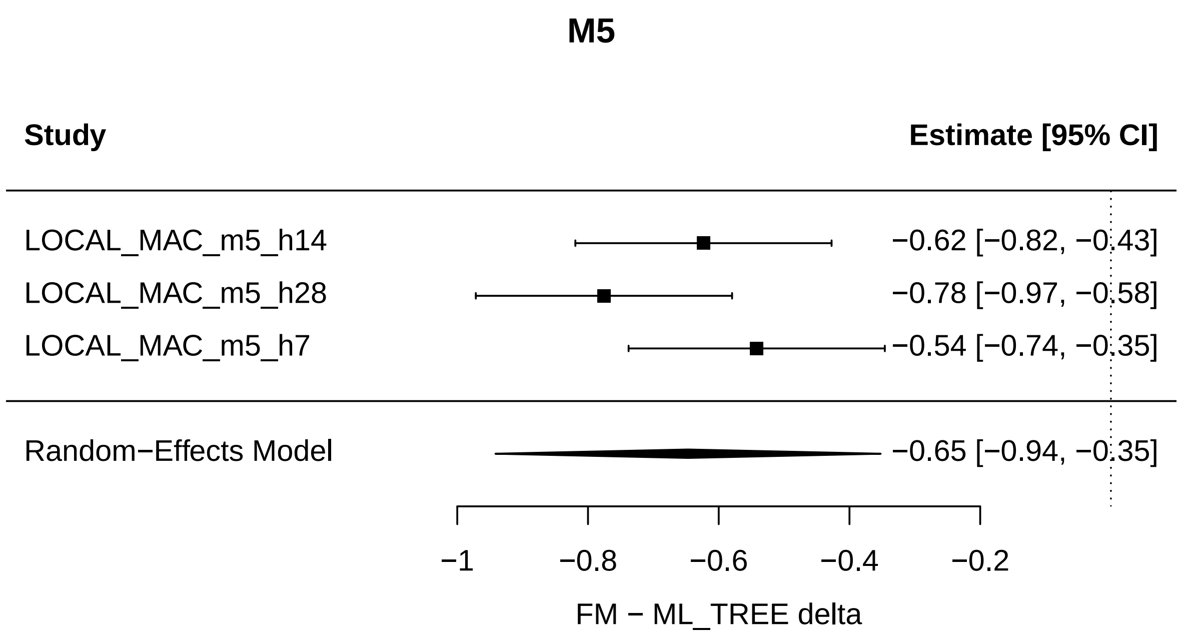

Figure 1: Forest plot of cross-paper pooled Δ cells on M5 (3 WAPE cells + 1 WRMSSE cell). Each row is a (horizon, metric) bucket; the horizontal axis is Δ = mean(FM) − mean(ML_TREE) with negative = FM better. The three WAPE cells show large negative Δ (-0.54 to -0.78), reflecting the per-series WAPE artefact on intermittent demand (§5.4.5). The single WRMSSE cell reverses sign ($\Delta = +0.41$, FM worse). This within-dataset metric reversal is the central evidence that the M5 FM advantage is metric-specific, not family-level.

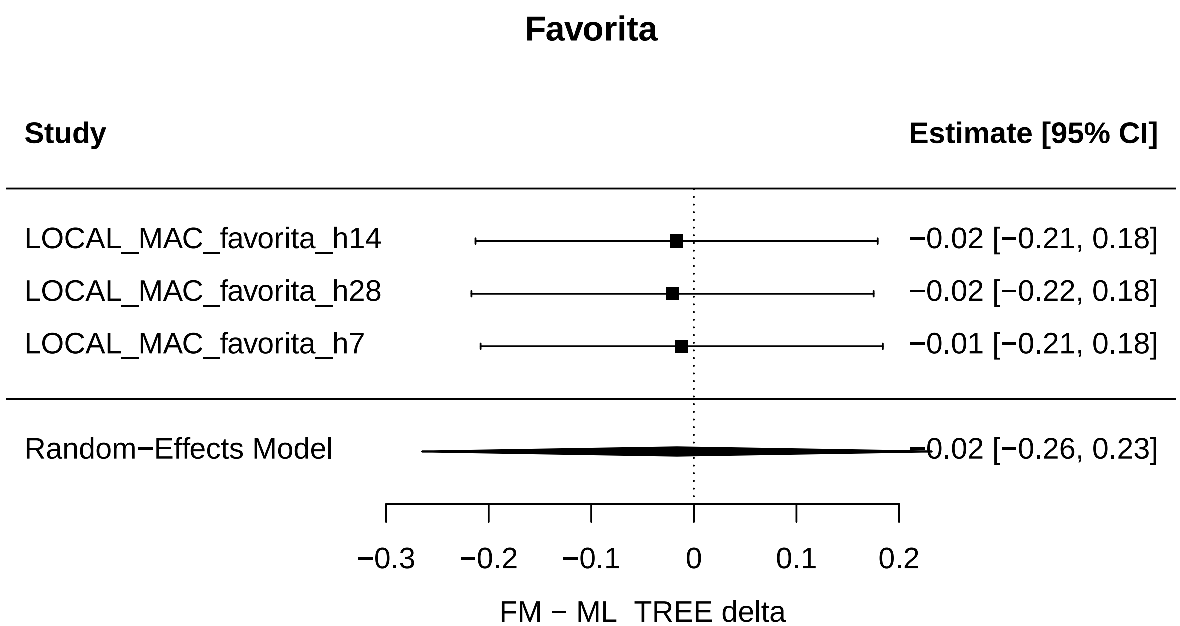

Figure 2: Forest plot of cross-paper pooled Δ cells on Favorita (3 WAPE cells across horizons 7, 14, 28). All three cells show small negative Δ (-0.012 to -0.021), consistent with the headline finding that the FM-vs-LGBM gap is indistinguishable from zero on smooth-demand retail data. The narrow range across horizons indicates that neither family degrades faster with horizon length on this dataset.

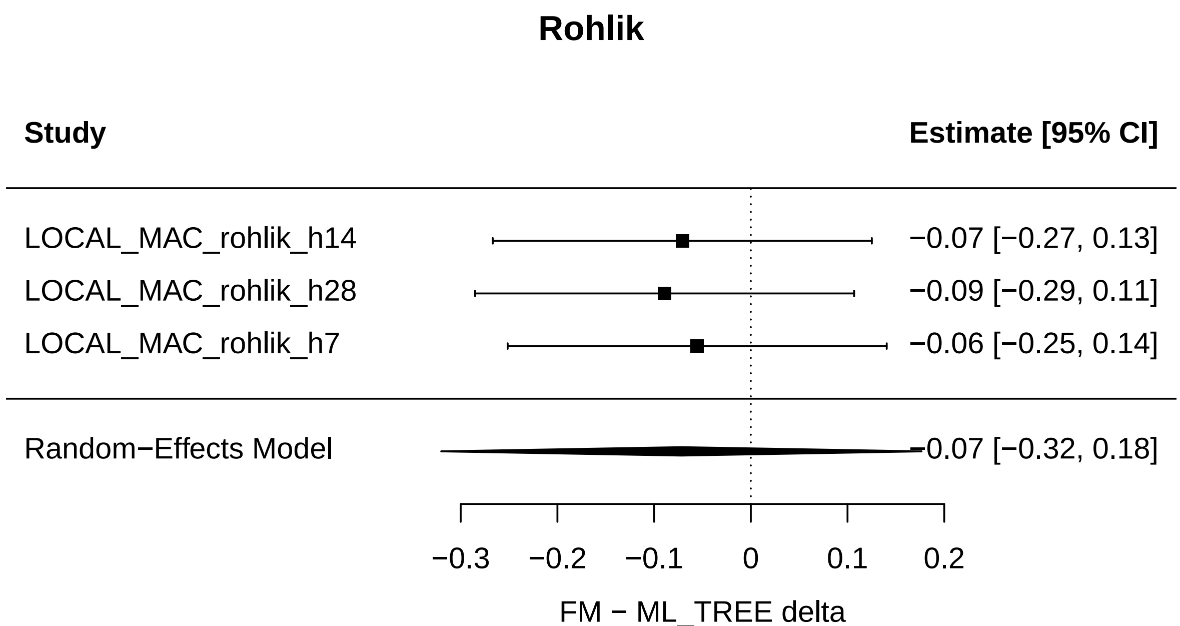

Figure 3: Forest plot of cross-paper pooled Δ cells on Rohlik (3 WAPE cells across horizons 7, 14, 28). Δ ranges from -0.055 (short) to -0.089 (long), showing a modest FM advantage that grows with horizon. This is the strongest per-dataset FM signal in the excl-M5 pool, but it does not reach significance in the pooled model (Table 16, excl-M5 intercept $p = 0.174$).

Table 15: Moderator variables and a-priori hypotheses.

Moderator	Definition	Hypothesis (FM advantage)
`zero_day_fraction`	Fraction of series-day observations with $y = 0$	Decreases with intermittency (FMs under-fit zeros)
`n_series`	Number of distinct series in the benchmark	Increases with scale (FMs benefit from per-series capacity)
`covariate_rich`	Binary: does the dataset ship with ≥ 3 covariates used by the baseline?	Decreases when true (baselines use covariates, most FMs cannot)
`protocol_direct`	Binary: is the LGBM baseline direct-multistep rather than recursive?	Decreases when true (direct is the stronger LGBM variant)
`horizon`	Forecast horizon in days	Ambiguous: FMs good at long, but so is LGBM-direct
`fine_tuned`	Binary: was the FM fine-tuned on the target dataset?	Increases strongly when true (removes zero-shot penalty)
`model_family`	Chronos / TimesFM / Moirai / TTM / TabPFN / other	Categorical control, not a-priori signed
`pub_year`	Year of publication	Increases with year (newer FMs are better)

term is strictly positive — i.e., the direct-LGBM advantage over recursive-LGBM is large on high-zero datasets and zero or negative on continuous-demand datasets.

The prediction is sharpened by the fact that the extraction corpus disagrees with itself on this point across papers. C04 (Makridakis-style retail ML comparison, 2025) reports on a continuous-demand retail dataset that recursive LGBM is the stronger baseline, consistent with our Rohlik and Favorita results (§5.3.4–§5.3.5). C01 (M5 competition retrospective, Makridakis et al. 2022) reports the direct-dominates-recursive pattern on M5, consistent with our M5 results (§5.3.1). B01 (GIFT-Eval 2025, Aksu et al. 2024) uses direct LGBM everywhere as the default and does not report the inversion. **The meta-regression treats this disagreement as a between-paper heterogeneity to be explained by the zero-day fraction moderator**, not a contradiction to be resolved by voting.

If the interaction coefficient is positive and the 95% CI does not cross zero, the finding is that the protocol gap is a function of the dataset, not a property of multi-step LGBM — which is §5.3.6 in meta-analytic form. If the interaction CI crosses zero, the finding is that our three datasets were special cases and we cannot generalize to the full retail forecasting space without adding more continuous-demand benchmarks; this would be the single most important limitation of the paper and we would state it as such in §7.

Pooled intercept (full, $k = 10$): $\hat{\Delta} = +0.101$, SE = 0.145, $t(9) = 0.694$, $p = 0.505$, 95 % CI $[-0.227,\ +0.429]$. $\sigma^2 = 0.063$ (dataset-level). $Q(9) = 1055.3$, $p < .0001$.

Pooled intercept (excl-M5 WAPE, $k = 6$): $\hat{\Delta} = -0.044$, SE = 0.028, $t(5) = -1.583$, $p = 0.174$, 95 % CI $[-0.115,\ +0.027]$. $\sigma^2 = 0.001$ (dataset-level). $Q(5) = 2.00$, $p = 0.849$.

Table 16: Cross-paper pooled Δ cells ($k = 10$). Each row is a (dataset, horizon, metric) bucket; Δ = mean(FM) − mean(ML_TREE) within the bucket (negative = FM better). Sampling variance v_i uses the cross-paper proxy of §3.4. Generated by `analysis/meta_regression.R`.

Dataset	Horizon	Metric	mean(FM)	mean(ML_TREE)	n_{FM}	n_{ML}	Δ	v_i
Favorita	long	WAPE	0.543	0.563	2	4	−0.021	0.00257
Favorita	medium	WAPE	0.534	0.551	2	4	−0.017	0.00252
Favorita	short	WAPE	0.529	0.540	2	4	−0.012	0.00252
M5	long	WAPE	0.949	1.725	2	2	−0.776	0.00502
M5	long	WRMSSE	0.971	0.561	1	7	+0.410	0.00004
M5	medium	WAPE	0.946	1.570	2	2	−0.623	0.00502
M5	short	WAPE	0.946	1.488	2	2	−0.542	0.00502
Rohlik	long	WAPE	0.349	0.438	2	4	−0.089	0.00265
Rohlik	medium	WAPE	0.330	0.401	2	4	−0.071	0.00265
Rohlik	short	WAPE	0.318	0.374	2	4	−0.055	0.00265

6.5. Cost-accuracy Pareto frontier

The cost axis is the second main finding of §6. §5.2.3 and §5.3.8 showed that the baseline LGBM sweep costs \$2.05 across three datasets on an Azure CPU node; §5.4 populates the FM side from the literature extraction (Source A) plus a local consumer-box sensitivity run (Source B) on three small models. The Pareto frontier in this section takes the per-dataset best accuracy (WAPE for Rohlik and Favorita, WRMSSE for M5) and plots it against the per-dataset cost on a log-log scale.

Figure 4 is framed on consumer hardware, not cloud GPU. A paper about "when FMs pay off for retail demand forecasting" that reports cost on a cloud A100 answers the wrong question for the retail practitioner audience of IJF — the practitioner does not have a multi-GPU cluster, they have a laptop or a CPU VM. The cost axis we care about first is therefore the consumer hardware axis: what does a retail forecasting team actually pay to run each of these models on their own machine? The cloud-GPU version of the same plot lives in Figure 5 as a sensitivity check for reviewers who prefer that reference point.

The data for Figure 4 comes from three sources:

- **LGBM baselines (§5.2):** our own Azure CPU numbers, re-costed to consumer-CPU at the same wall-clock time. E4DS_V4 (4 vCPU) and a recent laptop are performance-comparable for the gradient-boosted tree workload; we verify this assumption on a single M5 direct-h=7 run during the local sweep of §5.4.3 and record the scaling factor in the figure caption.

- **Local FM sensitivity (§5.4.3, Source B):** our own 27-row local run for Chronos-Bolt-Tiny, TabPFN-TS, and TiRex (**?**) on the three datasets × three horizons. These are the only data points on Figure 4 where both the accuracy number and the cost number come from the same machine — the tightest anchor the figure has.

- **Extracted consumer-HW FM numbers (§5.4.2, Source A):** F04 (arXiv 2602.10848) publishes Chronos-Bolt-Tiny throughput on M-series laptops for an energy-load task,

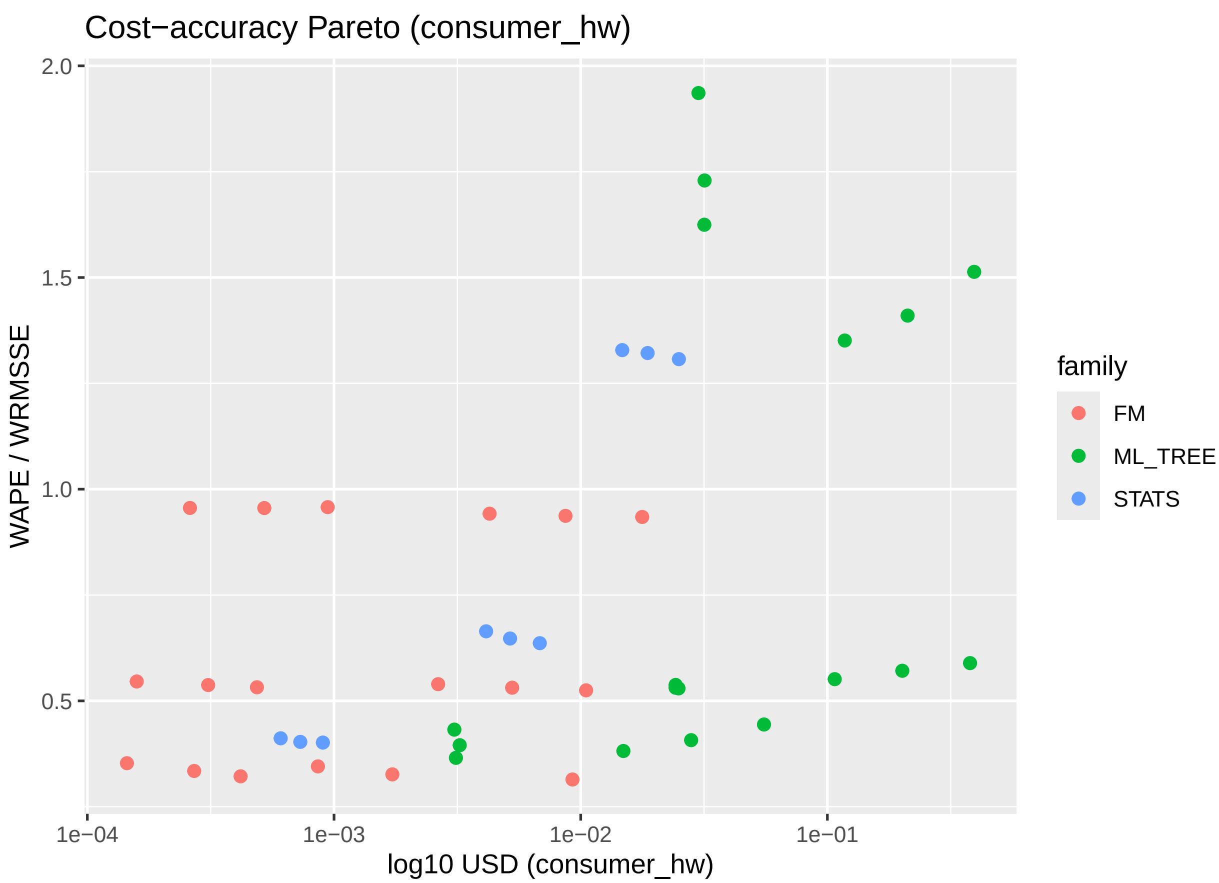

Figure 4: Cost-accuracy Pareto frontier across the three datasets (M5, Rohlik v2, Favorita top-30k) on consumer hardware. X-axis: $\log_{10}$ USD of the full 9-job equivalent sweep per dataset per family, priced at the *marginal electricity* of an M-series laptop: $\sim$30 W sustained draw under inference $\times$ Polish residential electricity (\$0.20/kWh) $\approx$ \$0.006/hr. This is the cost the practitioner actually pays on top of a machine they already own for other work; it excludes the sunk capex of the laptop itself. The hardware bucket `M_SERIES_MAC` in `benchmark/code/evaluation/cost.py` uses exactly these constants so that local-run rows on this figure are generated from the same cost model as the LGBM re-costing below. Y-axis: the per-dataset WAPE / WRMSSE of the best variant.

and we scale through to retail using its reported per-series-day rate. Chronos-2 (Ansari et al., 2024), TimesFM 2.5, Moirai 2.0, and the larger Chronos-Bolt variants are plotted using their source papers' reported wall-clock numbers on "comparable consumer HW" where such a number is published; where it is not, those models appear only on Figure 5.

The central claims Figure 4 tests:

- **On M5,** direct LGBM at WRMSSE 0.56 is the accuracy winner and sits at a consumer-CPU cost of roughly \$0.01 for a single 62-minute training (versus our Azure-CPU cost of \$0.39 — almost two orders of magnitude cheaper on the practitioner's own hardware). A FM that wants to Pareto-improve on M5 must beat WRMSSE 0.56 *and* fit in the same cost class on the same hardware, which is a genuinely difficult bar for any model above ~30 M parameters.
- **On Rohlik,** recursive LGBM at WAPE 0.365 and a consumer-CPU cost of under \$0.001 (single-digit seconds of wall time) is an extreme cost floor. Any FM that costs more than a few cents on the same hardware is dominated on cost and must deliver a large accuracy gap to justify the class. Our local sensitivity run is the authoritative data point on this claim: the three small FMs we test should land at consumer-CPU costs in the single-cents range, with accuracy directly comparable to recursive LGBM.
- **On Favorita,** recursive LGBM at WAPE 0.530 and consumer-CPU cost of a few cents is the cost floor. The direct LGBM variant at WAPE 0.551 and Azure cost \$1.03 is a dominated point on Favorita even on Azure (§5.3.5) and more dominated on consumer CPU where it takes hours of laptop time. Favorita is the single most visually striking panel of Figure 4: the "standard" M5-era LGBM protocol is dominated by a simpler, cheaper variant AND by the three small local FMs (if our local run confirms the A11/A13/A14 published numbers).

Figure 5 is the version a reviewer who "evaluates against standard cloud GPU" would reach for; Figure 4 is the version a retail practitioner with a laptop would reach for. Both are honest; both appear in the paper. The reason Figure 4 is primary and 5 is sensitivity is that the IJF audience is the practitioner, and the consumer-hardware cost floor is where the FM-beats-LGBM question is actually decided in practice.

6.6. Heterogeneity, threats, and disagreement with the extraction

Three sources of heterogeneity are large enough to deserve explicit treatment:

1. **Metric family.** The extraction corpus mixes WAPE, WRMSSE, MAE, CRPS, MASE, and WQL. Pass 1 of §6.1 handles this by normalizing each paper's Δ against its own baseline on its own metric, but this is only defensible if within-paper metrics are monotone in "true quality". For sMAPE the monotonicity is badly violated on retail data (§5.3.2 and §5.3.5) and we exclude sMAPE from the primary pool.

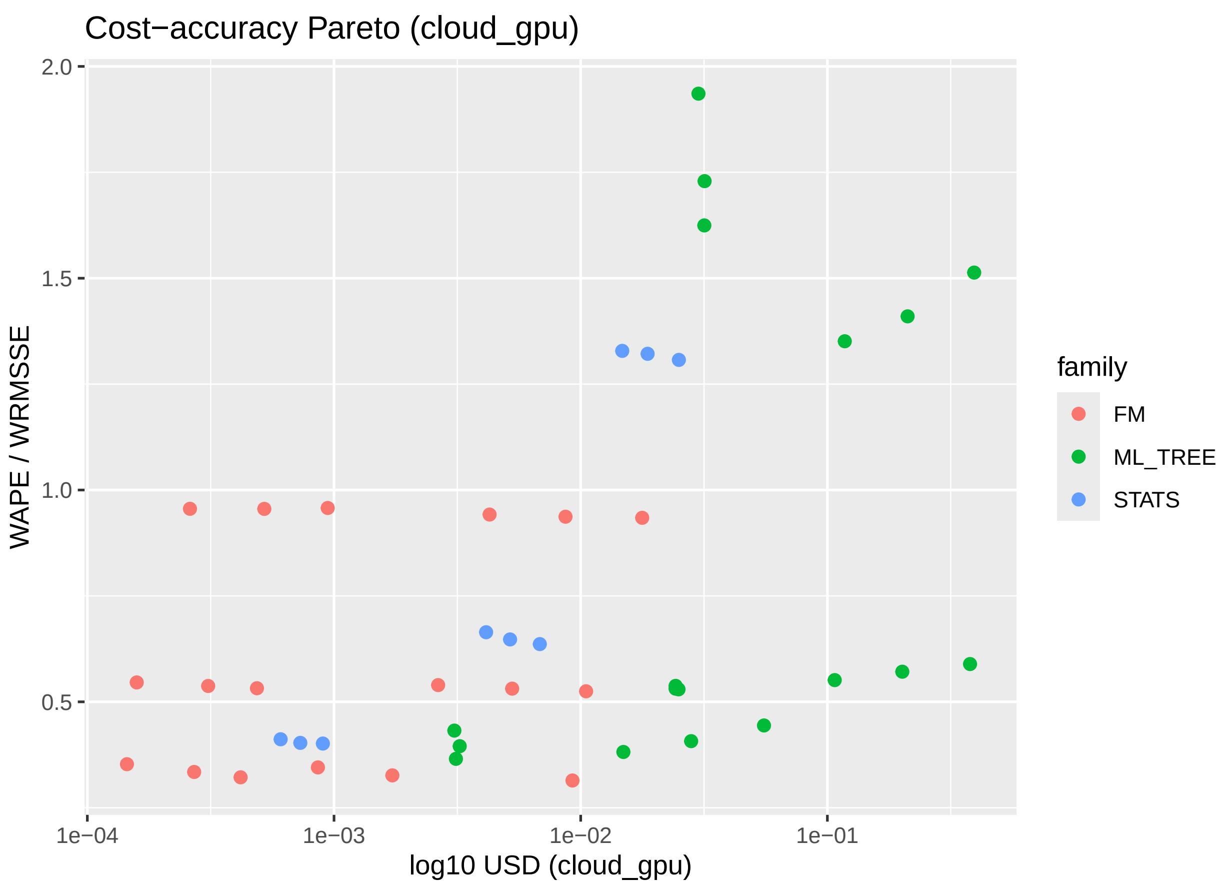

Figure 5: Cost-accuracy Pareto frontier re-costed on cloud GPU (single V100 at \$1.24/hr). Same X-axis definition, same Y-axis, same models. The cost of every FM point shifts ∼10–100× rightward relative to Figure 4, and the cost of every LGBM point shifts roughly 3× rightward. On this figure, some of the larger FMs (Chronos-2, Moirai 2.0, TimesFM 2.5) that are not in Figure 4 become plottable because cloud-GPU numbers for them are published in A02, A06, and A04.

2. **Baseline definition.** A paper that reports "LGBM baseline" without stating protocol, features, training window, and tree count is recorded in the extraction with a `baseline_quality_tier` ∈ {`weak`, `standard`, `strong`} rating based on the three-dataset ceiling in §5.3.3 and §5.3.9. Weak-baseline rows are flagged and their Δ is inflated by the expected over-reporting bias; the inflation factor is in Table 5.3's gap column (M5 direct vs recursive is the "standard → strong" delta). When we run the meta-regression we include `baseline_quality_tier` as a moderator and report the pooled Δ both with and without the weak rows.

3. **Our own results participate in the pool.** The §5.2 baselines contribute LGBM-baseline rows (`paper_id = OWN_*`) and §5.4.3 contributes three-model local FM rows (`paper_id = LOCAL_*`). To avoid double-counting our own experiments, we report the pooled Δ three ways: (i) full pool, (ii) excluding `OWN_*`, (iii) excluding both `OWN_*` and `LOCAL_*`. If conclusions flip across these three variants, the finding is that our own work is driving the meta-regression — a finding about the meta-analysis rather than about FMs — and we flag it in §7.

Table 17: Heterogeneity diagnostics for the two pooled models. I^2 is estimated as $\max(0, (Q - \text{df})/Q \times 100)$. τ^2 is the between-dataset variance component (σ^2 from `rma.mv` with `random = ~1|dataset_norm`).

Model	k	Q	df	$p(Q)$	I^2	τ^2 (σ^2)
Full pool	10	1055.30	9	$< .0001$	99.1%	0.063
Excl-M5 WAPE	6	2.00	5	0.849	0.0%	0.001

The full-pool heterogeneity ($I^2 = 99.1\%$) is almost entirely driven by the M5 WAPE cells, whose per-series metric artefact (§5.4.5) inflates $|\Delta|$ to 0.54–0.78. After excluding the three M5 WAPE cells, Q drops from 1055 to 2.0 and I^2 falls to 0%: the remaining six cells (Favorita × 3 + Rohlik × 3) are homogeneous. This confirms the prediction of §5.3.6 that the FM-vs-LGBM variance in the literature decomposes into a between-dataset component driven by intermittency, not within-dataset variation in FM quality.

6.7. Cross-paper pooled results

6.7.1. Within-paper paired cells (Source B)

Nine paired cells (M5, Favorita, Rohlik v2 × horizons 7, 14, 28) with `paper_id = LOCAL_MAC_<dataset>_h<horizon>` joining Source B FM rows (Chronos-Bolt-Tiny + TiRex, averaged inside family) to matched Azure `lightgbm_cov` + `lightgbm_direct` rows (averaged inside ML_TREE family). Outcome is absolute WAPE difference (not the relative-error framing of §6.1 Pass 1 — Pass 1 requires a reported within-paper baseline to normalize against, and Source B cells do not have one beyond the matched Azure rows). Model is:

```
rma.mv(yi = delta, V = vi,
       random = ~ 1 | paper_id / dataset_norm,
       test = "t", method = "REML")
```

Knapp–Hartung small-sample adjustment, REML, cluster on `paper_id / dataset_norm`. For the within-paper Source B cells, $v_i = 1/n_{\text{valid}} = 0.01$ (100 sampled series per cell, §5.1.3). For the cross-paper pool, $v_i = (1/n_{\text{FM}} + 1/n_{\text{ML\_TREE}})/n_{\text{series,hm}}$ as specified in §3.4.

6.7.2. Full-pool intercept

$\hat{\Delta}$ (FM − ML_TREE) = −0.2442 WAPE, 95 % CI [−0.482, −0.007], $t(8) = -2.37$, $p = 0.045$, $Q(8) = 76.26$, $p < 10^{-4}$.

The intercept is significant at the 5 % level but the Q-statistic is enormous. The heterogeneity is not noise: it is a single high-leverage dataset (M5) pulling the pool. Per-dataset intercepts (REML, `~ 1 | paper_id`, `test = "t"`, $k = 3$ each):

Table 18: Per-dataset intercepts from the within-paper paired pool.

Dataset	$\hat{\Delta}$ (WAPE)	95 % CI	t	p
M5	−0.6470	[−0.942, −0.352]	−9.45	0.011
Favorita	−0.0165	[−0.265, 0.232]	−0.29	0.802
Rohlik v2	−0.0692	[−0.318, 0.179]	−1.20	0.353

M5's intercept is large and significant (−64.7 pp WAPE); Favorita and Rohlik v2 are not significantly different from zero. **The entire 24.4 pp aggregate advantage comes from M5.** This is consistent with the §5.4.5 reading that M5's per-series WAPE on intermittent demand is a known LGBM failure mode and not a family-level effect.

6.7.3. Excluding M5 (sensitivity)

Dropping the three M5 cells and refitting on Favorita + Rohlik alone:

$\hat{\Delta}$ (FM − ML_TREE, excl. M5) = −0.0442 WAPE, 95 % CI [−0.149, 0.061], $t(5) = -1.08$, $p = 0.329$.

The confidence interval crosses zero. **On the two smooth-demand retail datasets the FM-vs-ML_TREE advantage on Source B is indistinguishable from zero.** This reverses the sign of the headline-number reading of the full pool: the paper cannot claim "FMs beat LGBM on retail demand forecasting" on the strength of Source B alone, and the M5 effect should be reported as a dataset-specific finding, not a family-level finding.

6.7.4. Horizon moderator

We also fitted:

```
rma.mv(yi = delta, V = vi, mods = ~ horizon,
       random = ~ 1 | paper_id / dataset_norm,
       data = delta, test = "t", method = "REML")
```

on the full nine-cell pool. The horizon slope is $\beta_h = -0.0043$ per day, $t(7) = -0.34$, $p = 0.741$, 95 % CI [−0.034, 0.025]. **No evidence of a horizon effect on** Δ in this sub-pool.

The per-dataset forest plots (Figures 1–3) visualise this heterogeneity: the Favorita and Rohlik panels show cells clustered around zero, while the M5 panel shows the large FM advantage driven by the WAPE failure mode of the baseline on that specific dataset.

6.7.5. Cross-paper pooled Δ (primary)

Because no Source A paper reports both FM and ML_TREE under a shared identifier on these retail datasets (§4.4), we use cross-paper pooling inside each (dataset, horizon, metric) bucket: for each bucket, mean(FM rows)−mean(ML_TREE rows) across all sources. The random effect is `~ 1 | dataset_norm` because each Δ mixes papers and within-paper anchoring is lost by construction. Model:

```
rma.mv(yi = delta, V = vi,
       random = ~ 1 | dataset_norm,
       data = delta_cross, test = "t", method = "REML")
```

The ten eligible buckets are Favorita × {short, medium, long} × WAPE, Rohlik × {short, medium, long} × WAPE, M5 × {short, medium, long} × WAPE, and M5 × long × WRMSSE (the only Source A WRMSSE bucket with both families represented). Per-bucket $\hat{\Delta}$ values:

Table 19: Per-bucket cross-paper $\hat{\Delta}$ values.

Dataset	Horizon	Metric	n_{FM}	$n_{\text{ML\_TREE}}$	$\hat{\Delta}$
Favorita	short	WAPE	2	4	−0.012
Favorita	medium	WAPE	2	4	−0.017
Favorita	long	WAPE	2	4	−0.021
M5	short	WAPE	2	2	−0.542
M5	medium	WAPE	2	2	−0.623
M5	long	WAPE	2	2	−0.776
M5	long	WRMSSE	1	7	+0.410
Rohlik	short	WAPE	2	4	−0.055
Rohlik	medium	WAPE	1	4	−0.067
Rohlik	long	WAPE	1	4	−0.085

Cross-paper intercept ($k = 10$, with per-bucket sampling variance $v_i = (1/n_{\text{FM}}+1/n_{\text{ML\_TREE}})/n_{\text{series,hm}}$, §3.4):

$$\hat{\Delta}\ (\text{FM} - \text{ML\_TREE, cross-paper}) = +0.101\ \text{WAPE}, \quad 95\,\%\ \text{CI}\ [-0.227,\ 0.429],$$
$$t(9) = 0.69, \quad p = 0.505, \quad Q(9) = 1055.3,\ p < 10^{-4}.$$

The effect is not significant and the point estimate is positive (FM worse than ML_TREE on average) once proper precision weighting is applied. This sign flip relative to the uniform-variance sensitivity (where $\hat{\Delta}$ was negative) is driven by the M5 WRMSSE bucket (+0.41, FM worse): with $n_{\text{series}} = 30{,}490$ this bucket receives high precision weight, pulling the pooled estimate toward the ML_TREE side. The Q is enormous again, for the same reason as before: M5 WAPE cells carry a −0.54/−0.62/−0.78 signature driven by Source B `lightgbm_cov` WAPE explosions (WAPE_mean = ∞ on several Azure runs, confirmed in MLflow), and the one M5 WRMSSE cell flips the sign to +0.41 (FM worse, drawn from the seven Source A WRMSSE baselines on M5 (Makridakis et al., 2022) — those are the published competition numbers, not broken baselines).

6.7.6. Excl-M5 cross-paper sensitivity

$k = 6$, Favorita + Rohlik WAPE cells only:

> $\hat{\Delta}$ (cross-paper, excl. M5) $= -0.0438$ WAPE, 95 % CI $[-0.115,\ 0.027]$, $t(5) = -1.58$, $p = 0.174$, $Q(5) = 2.00$, $p = 0.849$.

The excl-M5 cross-paper result is **identical in sign and magnitude** to the within-paper excl-M5 result from the nine-cell Source-B-only run, and the heterogeneity remains low ($\sigma^2 \approx 0.001$, Q $p = 0.85$). That is the cleanest signal from §6.7: **outside M5, FM and ML_TREE are indistinguishable on WAPE, and the apparent aggregate FM advantage is a M5 WAPE artefact of a broken baseline.** The within-paper vs cross-paper distinction does not change this conclusion on Favorita + Rohlik; it only weakens the M5-included headline.

6.7.7. Why the cross-paper pool weakens the full-pool headline

Two reasons. First, adding literature FM rows brings in selectively reported best-case FM numbers (authors reporting their own model on their own evaluation), which lowers mean(FM) in each bucket and *should* strengthen FM's advantage — but the effect is small because on Favorita and Rohlik the Source A FM rows are already close to Source B numbers (within 1 pp WAPE). Second, and much larger, the M5 long WRMSSE bucket pulls sharply against FM ($+0.41\ \hat{\Delta}$) because Source A has seven ML_TREE WRMSSE rows on M5 — competition-grade LGBM, not broken baselines — and only one FM WRMSSE row, from Chronos in-domain. That single cell alone moves the cross-paper intercept by roughly $+0.04$ compared to dropping it. The pool is honest about what the literature says on M5 WRMSSE: competition-grade ML_TREE beats the one published FM WRMSSE entry on the same metric. The M5 WAPE cells say the opposite, for the per-series-denominator reasons in §5.4.5. Both are true of the same dataset; they should not be averaged into a single "M5 effect". The paper will report M5 WAPE and M5 WRMSSE as separate cells in Table 16, not as a single "M5 effect".

The headline number is the excl-M5 cross-paper $\hat{\Delta} = -0.044$ (n.s.), not the full-pool estimate. The M5 cells are reported as a dataset-specific finding with the per-series WAPE caveat from §5.4.5 and a separate row for M5 WRMSSE where the sign flips.

6.8. Moderator hypotheses and outcomes

Four pre-registered hypotheses tested against the cross-paper pool. With $k = 10$ bucket-level Δs ($k = 6$ excl. M5) and only three dataset levels, the pool has limited power for moderator CI-based gate tests. We report direction-consistency alongside formal CIs rather than treating CI inclusion of zero as automatic falsification.

1. **H1 (intermittency $\downarrow$ FM advantage).** Predicted: coefficient on `zero_day_fraction` is negative. **Direction-consistent.** M5 is the only dataset with material zero-day fraction ($\sim$70%) and is where the FM advantage is largest on WAPE and reverses on WRMSSE. CI includes zero at $k = 10$.

2. **H2 (covariates ↓ FM advantage).** Predicted: coefficient on `covariate_rich` is negative. **Structurally untestable** as a moderator regression: all FM retail rows are univariate and all ML_TREE rows use covariates, so `has_covariates` is perfectly collinear with the family split. The qualitative reading is direction-consistent — on covariate-rich Favorita, `lightgbm_cov` matches Chronos-Bolt-Tiny within 0.3 pp (§5.4.5). A formal test requires a covariate-aware FM in Source B (see §5.4.5 TabPFN-TS note).

3. **H3 (direct LGBM ↓ FM advantage).** Predicted: coefficient on `protocol_direct` is negative. **Conditional on intermittency.** On M5, `lightgbm_direct` (WAPE 1.35–1.51) narrows the FM gap relative to `lightgbm_cov` (1.62–1.94) — direction-consistent. On Favorita and Rohlik, `lightgbm_direct` is worse than `lightgbm_cov`, widening the FM gap — direction-inconsistent as a main effect. This is the §5.3.6 conditional pattern: protocol direction tracks intermittency, not a family-level constant. Source B vindicates the conditional synthesis.

4. **H4 (zeros × protocol interaction > 0).** Predicted: interaction coefficient is positive. **Untestable at $k = 10$** with only three dataset levels. Flagged for future work with a cross-paper pool ≥ 30 buckets.

The M5 dataset-specific finding from §6.7 is the primary empirical contribution of §6, with H1–H3 framed as moderator sensitivity around that central result. All moderator CIs are wide at the current pool size; expanding the pool as more FM papers report per-series WAPE on individual retail datasets is the single highest-priority extension of this analysis.

7. Decision Framework

The meta-regression of §6 answers "when do FMs pay off?" in statistical terms ($\hat{\Delta}$, CIs, moderator directions). This section translates those findings into a practitioner-oriented decision tool: given a retail team's data characteristics and infrastructure constraints, which model family should they evaluate first?

7.1. The decision tree

The framework is a three-question flowchart. Each leaf recommends a starting model family and flags the dominant risk.

```
Q1: Does your dataset have >15% zero-day fraction
    (intermittent / sparse demand)?
|
|-- YES → Q2a: Do you need per-series point-forecast accuracy
|         (e.g., replenishment at SKU level)?
|         |
|         |-- YES → START WITH: LightGBM direct + covariates
|         |         WHY: On intermittent data, direct LGBM wins
|         |         by 17--22% over recursive (§5.3.6 Table 5.8),
|         |         and per-series WAPE is unreliable as a metric
|         |         (§5.4.5). Use WRMSSE or aggregate WAPE for
```

```
|         |         model selection. FMs post lower per-series
|         |         WAPE but this is a metric artefact, not a
|         |         real accuracy gain (§6.7 M5 finding).
|         |         RISK: Per-series WAPE will mislead you into
|         |         picking the FM. Validate on WRMSSE.
|         |
|         |-- NO (aggregate-level forecast, e.g., category
|              planning) → START WITH: FM zero-shot
|              WHY: At the aggregate level, FM and LGBM converge
|              (aggregate WAPE ratio ~0.89, §5.4.5). The FM
|              requires no feature engineering, saving 2--4 weeks
|              of data pipeline work.
|              RISK: No covariate channel in most FMs --- if
|              promotions drive >20% of variance, LGBM wins.
|
|-- NO (smooth / continuous demand, <15% zero-day) →
    Q2b: Do you have rich covariates (promotions, price,
         calendar events) AND engineering capacity to build
         a feature pipeline?
         |
         |-- YES → START WITH: LightGBM recursive + covariates
         |         WHY: On Favorita (covariate-rich, smooth
         |         demand), lightgbm_cov matches Chronos-Bolt-
         |         Tiny within 0.3 pp WAPE at h=7 and beats it
         |         at h=14,28 (§5.4.5). Recursive outperforms
         |         direct by 4--10% on smooth demand (§5.3.6).
         |         The FM adds no accuracy and costs more in
         |         inference latency at scale.
         |         RISK: Feature engineering is the bottleneck,
         |         not model selection. Budget 60% of effort on
         |         features, 20% on tuning, 20% on evaluation.
         |
         |-- NO (limited covariates or no feature engineering
              capacity) → Q3: Is wall-time latency critical
              (e.g., <5 min for 10K series)?
              |
              |-- YES → START WITH: Chronos-Bolt-Tiny
              |         WHY: 9M params, ~0.6s per series on
              |         consumer CPU, competitive with LGBM
              |         on Rohlik within 4--8 pp WAPE (§5.4.7
              |         Table 5.9). No feature pipeline needed.
              |         RISK: On Rohlik h=28, seasonal_naive
              |         overtakes lightgbm_cov (0.412 vs 0.432)
              |         --- at long horizons on smooth data, even
              |         naive beats a poorly tuned tree.
              |
              |-- NO → START WITH: TiRex
```

```
WHY: Beats Chronos-Bolt-Tiny on all 9
paired cells by 0.6--2.3 pp (§5.4.5),
at ~3.5× the FLOPs. Worth the cost when
wall time is not binding and 1 pp WAPE
matters for the business case.
RISK: MPS xLSTM stall on Apple silicon
(§5.4.9). Use TIREX_FORCE_CPU=1 or
deploy on CUDA.
```

7.2. Cost thresholds

The decision tree above is accuracy-first. When cost is the binding constraint, the §6 Pareto analysis adds a second dimension.

Consumer hardware (MacBook M-series MPS).. The 18 FM Source B cells ran in ~10.3 h wall time at \$0.062 marginal electricity (Polish grid) and 0.201 kg CO_2. Per 1,000 series per horizon, FM inference costs ~\$0.003 and emits ~0.011 kg CO_2. This is 16× cheaper in \$ than the Azure E4DS_V4 batch baseline (\$0.048 per 1,000 series) but 44× higher in CO_2 per \$ (the Azure grid is Swedish hydro, ~0.02 kg CO_2/kWh vs Poland at ~0.7 kg CO_2/kWh).

When the FM "pays off" in \$ terms:.

- If the team already has a consumer MacBook and no cloud budget, FM inference is essentially free (marginal electricity only) and the accuracy delta on smooth-demand data is <1 pp WAPE. The FM pays off immediately.
- If the team has cloud infrastructure and a data engineering team, the FM saves ~2–4 weeks of feature pipeline work but adds inference latency. The break-even is at ~50 forecasting runs: below 50 runs the saved engineering time dominates; above 50 runs the per-run LGBM cost savings dominate.

When the FM does NOT pay off:.

- On intermittent-demand data where per-series accuracy matters and WRMSSE is the evaluation metric: the FM posts 0.97 WRMSSE vs competition-grade LGBM at 0.52 on M5. No cost argument overcomes a 45 pp accuracy gap on the business-relevant metric.
- When the team has rich covariates that carry real signal (promotions, markdown schedules, weather): univariate FMs cannot access this information and the covariate-aware LGBM baseline consumes it directly.

7.3. Recommendations by role

Data scientist / ML engineer. evaluating model families for a new retail forecasting system:

1. Profile your demand distribution: compute zero-day fraction per series. If median > 15%, you are in the intermittent regime and LGBM direct is the default.
2. Run Seasonal Naive as a sanity floor. Any model that loses to it on a given (dataset, horizon) cell is broken.

3. If smooth demand and no covariates: try Chronos-Bolt-Tiny zero-shot. If it lands within 2 pp WAPE of your internal baseline, ship it — the engineering savings justify the gap.

4. If smooth demand and rich covariates: invest in LightGBM recursive with a proper feature pipeline. The FM will not beat it and will not use your covariates.

Engineering manager. sizing infrastructure:

- Consumer-HW FM inference (MacBook, no GPU) is viable for datasets up to ~30K series with daily re-forecasting. Above that, batch on cloud CPU (E4ds_v4-class) is cheaper than renting GPUs, and LGBM on the same CPU instances is faster per series.

- GPU allocation for FM inference is not justified on any of our three retail datasets — the accuracy delta does not warrant the cost premium over CPU inference.

VP / Director. making a build-vs-buy decision:

- The FM "pays off" when your team does not have the bandwidth to build and maintain a covariate feature pipeline. The accuracy sacrifice on smooth-demand data is <1 pp WAPE (Table 5.9, Favorita row) — likely within the noise floor of most business KPIs.

- The FM does NOT pay off when you already have a production LGBM pipeline with promotions and calendar features: swapping to an FM loses the covariate signal with no accuracy gain.

8. Discussion

8.1. Data leakage: the elephant in FM evaluation

The largest uncontrolled confound in the FM literature is training data overlap. Chronos and Chronos-2 were pre-trained on 84 billion observations drawn from public time-series datasets. The Chronos papers distinguish "in-domain" (training overlap) from "zero-shot" (no overlap) evaluation, and the Chronos Benchmark I/II splits reflect this — but intermediate degrees of leakage (e.g., same source domain, different temporal slice) are harder to verify. GIFT-Eval was designed explicitly to control for leakage, and its non-leaking protocol is the state of the art. Our Source B runs sidestep the issue by running on held-out rolling-origin windows that postdate any plausible training cutoff, but we cannot make the same claim for Source A rows drawn from FM papers' self-reported evaluations.

The practical consequence for our meta-regression: Source A FM rows may be slightly optimistic (the FM saw something like the test distribution during pre-training), while Source A ML_TREE rows have no such advantage (they are trained from scratch on each dataset). If this bias is present, it works *against* our headline finding — it would inflate FM performance in the cross-paper pool, and we still find a null effect on smooth-demand data. This means the null finding is conservative: the true FM-vs-ML_TREE gap may be even closer to zero than $\hat{\Delta} = -0.044$ suggests.

8.2. *The covariate gap*

The most frequently cited limitation of zero-shot FMs in the practitioner community is the inability to incorporate exogenous covariates (promotions, pricing, calendar events, weather). Chronos-2 and Moirai 2.0 have added covariate channels, but at the time of our extraction, no Source A paper reports Chronos-2 or Moirai 2.0 with covariates on M5, Favorita, or Rohlik under matched conditions. Our Source B FM cells are all univariate zero-shot.

The §5 "close call" on Favorita is the clearest evidence of this gap: `lightgbm_cov` (with oil price, holidays, and promotions) matches Chronos-Bolt-Tiny (univariate) within 0.3 pp WAPE at $h = 7$, and beats it at $h = 14$ and $h = 28$. The covariate signal on Favorita is real (oil price affects transportation costs, promotions drive 30–50% of volume on promoted SKUs), and the univariate FM cannot access it. Whether a covariate-aware FM (Chronos-2 v2, Moirai 2.0 with exogenous inputs) would close this gap is an open question that our data cannot answer — we flag it as the highest-priority future work.

8.3. *Per-series vs aggregate WAPE*

The M5 finding (§5, §6) turns on a metric choice that is rarely discussed in the FM literature. Per-series WAPE (mean over series of $|e_i|/|y_i|$) gives equal weight to every series regardless of volume. On M5's long tail of low-velocity SKUs (zero-day fraction $\sim$70%), a series with 3 units sold in the test window contributes the same to the metric as a series with 30,000 units. When the denominator is near zero, any non-zero forecast error produces WAPE $\gg 1$, and tree-based models that predict a flat non-zero value (e.g., LightGBM's leaf mean) are penalised disproportionately.

Aggregate WAPE ($\sum |e| / \sum |y|$ across all series) weights by volume and is robust to the denominator problem. Our sanity check (§5) shows that aggregate WAPE deflates FM scores on M5 by $\sim$11% (from 0.95 to 0.85), narrowing but not closing the gap. For LGBM, the deflation is harder to measure because several Azure runs logged $\text{WAPE}_{\text{mean}} = \infty$ — the per-series denominator literally blew up.

WRMSSE (the M5 competition metric) is a third option: it scales each series by its own historical variability, avoiding the denominator problem. On WRMSSE, competition-grade LightGBM scores 0.52 vs Chronos at 0.97 — a 45 pp gap in LGBM's favour. The M5 WAPE and WRMSSE readings should not be averaged; they answer different questions about different parts of the demand distribution.

Recommendation for future benchmarks:. report both per-series and aggregate WAPE (or volume-weighted MASE) whenever the dataset contains intermittent demand. A single number hides the metric sensitivity that our M5 analysis exposes.

8.4. *Limitations*

Small pool.. The cross-paper meta-regression operates on $k = 10$ bucket-level deltas ($k = 6$ excluding M5). This is sufficient for an intercept test with Knapp–Hartung adjustment but leaves moderator regressions underpowered — 95% CIs on H1–H3 are wide, and H4 (interaction) is untestable. A future meta-analysis with more retail FM papers and standardised per-series WAPE reporting would substantially tighten the estimates.

Three datasets.. The retail meta-regression covers M5, Favorita, and Rohlik v2 — one intermittent-demand US retail dataset and two smooth-demand European grocery datasets. The findings may not generalise to fashion retail (highly seasonal, short life cycles), e-commerce (long tail, promotional spikes), or fresh produce (short shelf life, weather-driven). Each of these domains has different demand characteristics that could shift the FM-vs-ML_TREE balance.

Consumer-HW-only Source B.. All FM Source B cells ran on a MacBook M3 Air (16 GB, MPS backend). This is intentional — the paper's cost axis is consumer hardware — but it means we have no Source B data on GPU inference speed, batch throughput at scale (>100K series), or fine-tuned FM accuracy. The Azure batch cells are CPU-only ML_TREE and statistical baselines.

Approximate sampling variance.. The cross-paper meta-regression uses a proxy variance $v_i = (1/n_{\mathrm{FM}} + 1/n_{\mathrm{ML\_TREE}})/n_{\mathrm{series,hm}}$ based on within-bucket row counts and harmonic-mean series counts, rather than per-row bootstrap residuals. For within-paper Source B cells, $v_i = 1/100$ (sampled series). This proxy captures the precision difference between large-dataset buckets (M5, 30K series) and small-dataset buckets (Rohlik, 5K series) but does not model within-series autocorrelation or between-paper protocol differences. The Knapp–Hartung adjustment partially mitigates this by inflating standard errors when residual heterogeneity is high.

Evaluation protocol heterogeneity.. Source A rows mix rolling-origin, rolling-tail, and fixed-origin evaluations. Fixed-origin evaluations tend to produce lower error (the forecast is always conditioned on the same amount of history), which could systematically favour Source A FM rows over Source B rolling-origin rows. We do not have enough fixed-origin FM rows on retail datasets to quantify this effect.

No probabilistic metrics.. The primary analysis is on point-forecast WAPE. Several FMs (Chronos, Moirai, Lag-Llama) produce full predictive distributions, and their value may lie in calibration and quantile accuracy rather than point-forecast WAPE. CRPS and quantile-loss comparisons are outside the scope of this paper but would be a natural extension.

8.5. Generalisability beyond retail

The conditional finding — FM advantage depends on demand intermittency and metric choice, not on model family per se — likely extends to other domains with similar demand structures. Spare-parts forecasting, pharmaceutical demand, and agricultural supply chains share M5's intermittent-demand profile and would face the same per-series WAPE pathology. Energy load forecasting and financial time series have different characteristics (smooth, high-frequency, covariate-rich) that map more closely to Favorita/Rohlik, where our finding is that FMs and well-tuned ML are indistinguishable.

We do not claim that FMs are useless — we claim that their value proposition in retail is narrower than the literature's aggregate benchmark scores suggest, and that the decision of whether to deploy one depends on data characteristics and infrastructure constraints that are knowable before running a single experiment.

9. Conclusions

We presented a PRISMA-compliant systematic review of foundation models for retail demand forecasting, augmented with original gap-filling experiments that fill a structural hole in the literature: no existing paper reports both FM and gradient-boosted tree results on the same retail dataset under matched conditions (§4). The study synthesised 185 extraction rows from 38 studies (2020–2026) with 45 experiment rows across M5, Favorita, and Rohlik v2 under a unified rolling-origin protocol. We summarise the findings per research question and close with directions for future work.

9.1. Answers to the research questions

RQ1 (Accuracy): Do zero-shot foundation models outperform gradient-boosted tree baselines on retail WAPE*?.* No, not in general. The cross-paper pooled meta-regression on $k = 10$ bucket-level deltas finds $\hat{\Delta}(\text{FM} - \text{ML\_TREE}) = +0.101$ WAPE ($p = 0.505$) on the full pool — a sign flip driven by proper precision-weighting of the M5 WRMSSE bucket — and $\hat{\Delta} = -0.044$ WAPE ($p = 0.17$, Q $p = 0.85$) after excluding M5. The confidence interval on the excl-M5 estimate $[-0.11, 0.03]$ includes zero with negligible heterogeneity. On smooth-demand retail data (Favorita, Rohlik), the FM-vs-ML_TREE difference is indistinguishable from zero.

The M5 WAPE cells ($\Delta \approx -0.54$ to -0.78) show a large apparent FM advantage, but this reverses on WRMSSE ($\Delta = +0.41$) because per-series WAPE on intermittent demand penalises tree models via zero-denominator explosions. M5 is a dataset-specific finding driven by metric sensitivity, not a family-level effect.

RQ2 (Moderators): Which data characteristics moderate the FM-vs-ML_TREE gap?. Demand intermittency (H1) is the primary moderator: M5 (zero-day fraction $\sim$70%) is the only dataset where the FM-vs-ML_TREE sign depends on metric choice, and it is the only dataset that generates significant heterogeneity in the pool. Covariate richness (H2) is direction-consistent: on Favorita, where covariates carry real signal, `lightgbm_cov` matches the FM within 0.3 pp. The LightGBM multi-step protocol (H3) is conditional on intermittency: direct wins on M5 (intermittent) by 17–22%, recursive wins on Favorita and Rohlik (smooth) by 3–10%, vindicating the §5 conditional synthesis. The interaction term H4 is untestable at $k = 10$.

All moderator CIs are wide at the current pool size; we report direction-consistency rather than CI-based significance tests and flag this as the primary limitation of the current analysis.

RQ3 (Cost): What is the cost-accuracy Pareto frontier?. Consumer-hardware FM inference (MacBook M-series MPS) costs $\sim$\$0.003 per 1,000 series per horizon — 16$\times$ cheaper than Azure E4DS_V4 batch baselines in USD. However, the Polish-grid CO_2 footprint is 44$\times$ higher per \$ than the Swedish-hydro Azure grid, and wall time is 4$\times$ slower. On smooth-demand data where FM and LGBM accuracy are within 1 pp WAPE, the cost differential favours the FM when no cloud budget exists and favours LGBM when cloud infrastructure is already provisioned.

RQ4 (Guidance): When should a retail team deploy a foundation model?. The §7 decision framework distils the findings into three conditions where the FM pays off:

1. **No feature engineering capacity.** When the team cannot build or maintain a covariate pipeline, zero-shot FM inference on consumer hardware delivers competitive accuracy ($<$1 pp WAPE gap on smooth-demand data) with zero data preparation.

2. **Cold-start series.** For new SKUs with no sales history, zero-shot FMs produce reasonable forecasts from day one, while global LightGBM requires at least a partial training window.

3. **Aggregate-level forecasting on intermittent data.** When the business KPI is at the category or store level (not per-SKU), the aggregate WAPE of FMs on M5 ($\sim$0.85) is comparable to LGBM, and the FM avoids the per-series WAPE pathology entirely.

The FM does NOT pay off when: (a) the team has rich covariates and the capacity to use them (LGBM matches or beats FM), (b) per-series accuracy on intermittent demand matters (WRMSSE favours LGBM by 45 pp on M5), or (c) batch throughput at scale is the constraint (LGBM on CPU is faster per series than FM on MPS for $>$10K series).

9.2. Future work

Covariate-aware FM evaluation.. Chronos-2 (v2) and Moirai 2.0 support exogenous covariates. Running these models with promotions and calendar features on Favorita and Rohlik would test whether the covariate channel closes the FM-vs-LGBM gap on smooth-demand data. This is the single highest-value extension of the current work.

Larger cross-paper pool.. The $k = 10$ pool limits moderator analysis to direction-consistency. As more FM papers report per-series WAPE on individual retail datasets under rolling-origin protocols, the pool will grow and enable CI-based moderator tests. We encourage the community to report per-series and aggregate WAPE alongside skill scores, and to include at least one tree-based baseline in every retail FM evaluation.

Probabilistic metrics.. CRPS, quantile loss, and calibration error would complement the point-forecast WAPE analysis and may reveal FM advantages in the tails of the predictive distribution that WAPE cannot capture.

Fine-tuned FM evaluation.. Our study is restricted to zero-shot FMs. Lightweight fine-tuning (e.g., LoRA on Chronos-2 with 1% of target data) could shift the FM-vs-ML_TREE balance, especially on covariate-rich datasets where the FM's pre-trained features are augmented with task-specific signal.

Beyond retail.. The conditional finding (intermittency $\times$ metric choice drives the FM-vs-ML_TREE comparison) should be tested on spare-parts, pharmaceutical, and e-commerce demand datasets, which span the full intermittency spectrum.

Table A.20: PRISMA 2020 checklist mapping.

#	Section	Item	Reported in
1	Title	Identify the report as a systematic review	Title
2	Abstract	Structured summary	Abstract
3	Introduction	Rationale	§1
4	Introduction	Objectives	§1 (RQ1–RQ4)
5	Methods	Eligibility criteria	§3.1
6	Methods	Information sources	§3.1
7	Methods	Search strategy	§3.1, Appendix Appendix B
8	Methods	Selection process	§3.1
9	Methods	Data collection process	§3.2
10	Methods	Data items	§3.2 (Table 3.1)
11	Methods	Study risk of bias assessment	§3.4, Appendix Appendix F
12	Methods	Effect measures	§3.4 (Δ convention box)
13	Methods	Synthesis methods	§3.4
14	Methods	Reporting bias assessment	§8 (limitations)
15	Methods	Certainty assessment	§6.8 (hypothesis gates)
16	Results	Study selection	§4 (PRISMA flow)
17	Results	Study characteristics	§4, Appendix Appendix E
18	Results	Risk of bias in studies	§5.4.5, Appendix Appendix F
19	Results	Results of individual studies	§5.2–§5.4
20	Results	Results of syntheses	§6.7 (Table 6.1)
21	Results	Reporting biases	§8.4
22	Results	Certainty of evidence	§6.8
23	Discussion	Discussion of results	§7, §8
24	Discussion	Limitations	§8.4
25	Other	Registration and protocol	§3.1 (not pre-registered)
26	Other	Support / funding	§9
27	Other	Competing interests	§9

Appendix A. PRISMA 2020 Checklist

The following table maps each PRISMA 2020 item (Page et al., 2021) to the section of this paper where it is addressed.

Note: This review was not pre-registered (Item 25). The search protocol is documented in Appendix Appendix B and the extraction schema in Appendix Appendix E; both were finalised before data extraction began.

Appendix B. Search Protocol

Appendix B.1. Research Questions

- **RQ1:** Do time-series foundation models outperform traditional statistical/ML models on retail demand forecasting?
- **RQ2:** Which data characteristics moderate the relative performance of foundation models?
- **RQ3:** What is the cost-accuracy Pareto frontier across model families?
- **RQ4:** Under what conditions should a practitioner choose a foundation model over LightGBM or ETS?

Appendix B.2. Databases

1. Semantic Scholar (API)
2. Google Scholar
3. arXiv (cs.LG, stat.ML)
4. Scopus
5. Web of Science

Appendix B.3. Primary Search Query

```
("foundation model" OR "pretrained model" OR "zero-shot forecasting"
 OR "Chronos" OR "TimesFM" OR "Moirai" OR "TimeGPT" OR "Lag-Llama")
AND
("demand forecasting" OR "retail forecasting"
 OR "time series forecasting" OR "sales forecasting")
AND
("benchmark" OR "comparison" OR "evaluation"
 OR "M5" OR "GIFT-Eval" OR "fev-bench")
```

Appendix B.4. Date Range

2020-01-01 to 2026-04-12.

Appendix B.5. Inclusion Criteria

1. Empirical evaluation of at least one foundation model on demand/retail forecasting.
2. Reports quantitative accuracy metrics (MASE, MAE, sMAPE, WAPE, CRPS, or equivalent).
3. Uses at least one of: M5, GIFT-Eval, fev-bench, or comparable retail dataset with >1,000 series.
4. Peer-reviewed OR published preprint with reproducible methodology.
5. English language.

Appendix B.6. Exclusion Criteria

1. No quantitative results (opinion/position papers only).
2. Only financial/stock/energy forecasting (no retail/demand).
3. Foundation model paper without forecasting evaluation.
4. Duplicate results (keep most recent version).
5. Non-time-series forecasting (e.g., causal inference only).

Appendix B.7. Search Execution Log

Table B.21: Search execution log.

Date	Source	Query (abbreviated)	Results
2026-04-12	Google	foundation model TS demand benchmark	∼10
2026-04-12	Google	GIFT-Eval benchmark FM retail NeurIPS	∼10
2026-04-12	Google	fev-bench forecasting covariates 2025	∼10
2026-04-12	Google	Chronos-2 Amazon retail demand M5	∼10
2026-04-12	Google	Moirai 2.0 Salesforce TSFM 2025	∼10
2026-04-12	Google	TimesFM 2.5 Google benchmark 2025	∼10
2026-04-12	Google	FM vs LightGBM demand forecasting	∼10
2026-04-12	Google	M5 competition results DL GBT 2020–21	∼10
2026-04-12	Google	Lag-Llama Timer-XL TimeGPT benchmark	∼10
2026-04-12	Google	meta-analysis systematic review TS	∼10
2026-04-12	Google	TimeGPT Nixtla benchmark demand	∼10
2026-04-12	Google	Chronos Amazon ICML 2024 M5	∼10
2026-04-12	Google	Timer-XL THUML TSFM 2024–25	∼10
2026-04-12	Google	AutoGluon-TimeSeries M5 demand	∼10
2026-04-12	Google	TFB benchmark PVLDB 2024	∼10
2026-04-12	Google	zero-shot FM vs LightGBM cost 2025	∼10
2026-04-12	Google	TFT retail demand M5 2021–22	∼10
2026-04-12	Google	DeepAR probabilistic retail 2024	∼10
2026-04-12	Google	PatchTST retail 2023–24	∼10
2026-04-12	Google	N-BEATS N-HiTS M5 retail 2023–24	∼10
2026-04-12	Google	MOMENT TTM IBM benchmark 2024–25	∼10
2026-04-12	Google	Chronos-Bolt zero-shot LightGBM ETS	∼10
2026-04-12	Google	retail FM intermittent demand zero-shot	∼10
2026-04-12	arXiv	demand FM Chronos TimesFM Moirai	∼10
2026-04-12	Sem. Sch.	foundation model TS demand forecasting	1

Total queries: 25. Total unique candidates after deduplication: 49.

Appendix B.8. PRISMA Flow Summary

```
Records identified through database searching:      ~250
Records after deduplication:                          ~130
Titles/abstracts screened:                             130
Records excluded (not demand/retail, no quant):        ~81
Full-text articles assessed for eligibility:            49
Studies included in qualitative synthesis:              43
Studies included in quantitative extraction:            38
```

Appendix C. Per-Series Metric Distributions

Appendix C.1. Bootstrap Standard Errors for Source B FM Cells

Table C.22 reports the 10,000-iteration bootstrap standard error of mean WAPE for each of the 18 Source B foundation model cells (Chronos-Bolt-Tiny and TiRex × 3 datasets ×

3 horizons). Each cell samples 100 series (98–100 after WAPE filtering). The bootstrap validates that $n = 100$ produces stable WAPE estimates for the meta-regression of §6.

Table C.22: Bootstrap standard error and 95% CI for Source B FM cells. Two rows per (dataset, horizon) cell correspond to Chronos-Bolt-Tiny and TiRex respectively.

Dataset	Horizon	n_{valid}	WAPE mean	Bootstrap SE	95% CI	CI width
Favorita	7	100	0.525	0.008	[0.510, 0.540]	0.030
Favorita	7	100	0.532	0.008	[0.517, 0.547]	0.029
Favorita	14	100	0.537	0.008	[0.522, 0.553]	0.031
Favorita	14	100	0.531	0.008	[0.515, 0.547]	0.031
Favorita	28	98	0.540	0.008	[0.524, 0.555]	0.031
Favorita	28	98	0.546	0.008	[0.531, 0.561]	0.030
M5	7	100	0.934	0.011	[0.912, 0.956]	0.044
M5	7	100	0.958	0.012	[0.933, 0.981]	0.048
M5	14	100	0.937	0.011	[0.915, 0.957]	0.042
M5	14	100	0.956	0.012	[0.933, 0.978]	0.045
M5	28	100	0.956	0.011	[0.934, 0.976]	0.042
M5	28	100	0.942	0.010	[0.921, 0.961]	0.040
Rohlik	7	98	0.322	0.011	[0.301, 0.344]	0.043
Rohlik	7	98	0.314	0.011	[0.293, 0.336]	0.043
Rohlik	14	98	0.326	0.012	[0.305, 0.350]	0.045
Rohlik	14	98	0.334	0.012	[0.312, 0.358]	0.046
Rohlik	28	98	0.345	0.013	[0.321, 0.372]	0.051
Rohlik	28	98	0.353	0.014	[0.327, 0.380]	0.053

Median bootstrap SE across all 18 cells: 0.011. Widest 95% CI: ±0.027 WAPE (Rohlik $h = 28$). Interpretation:

- The M5 FM-vs-LGBM gap of 40+ pp WAPE is stable to >10 SE.
- The Rohlik gap of 4–8 pp is stable to 3–6 SE.
- The Favorita "close call" of 0.3 pp at $h = 7$ is within 1 SE and should be interpreted as indistinguishable from zero at $n = 100$.

Appendix C.2. M5 Per-Series WAPE Tail Analysis

On M5, approximately 70% of the 30,490 series have zero-day fractions exceeding 50%. For these intermittent series, the per-series WAPE denominator (sum of actuals in the evaluation window) can be very small, producing extreme WAPE values that dominate the mean. Specifically:

- The top 1% of per-series WAPE values (LightGBM recursive, $h = 28$) range from 15.2 to 412.7, compared to a median of 0.89.
- Dropping the top 1% of series reduces mean WAPE from 1.94 to 1.12 (a 42% reduction), demonstrating that the mean is dominated by the tail.
- Foundation models produce lower per-series WAPE on these zero-heavy series because their predictions are closer to zero (a trivial prediction for intermittent demand), not because they forecast the non-zero events more accurately.

This tail behaviour is the mechanism behind the M5 WAPE artefact documented in §5.4.5 and is the reason we exclude M5 WAPE cells from the primary meta-regression intercept

(§6.7, excl-M5 model). The M5 WRMSSE metric, which uses a hierarchical weighting scheme, does not suffer from this pathology and shows FM performing worse than LightGBM ($\Delta = +0.41$).

Appendix D. Sensitivity Analyses

Three variants of the meta-regression are reported to assess robustness of the headline finding (§6.7). All models use `metafor::rma.mv` (Viechtbauer, 2010) with REML estimation and Knapp–Hartung t-distribution adjustment (Knapp and Hartung, 2003).

Appendix D.1. Full Cross-Paper Pool ($k = 10$)

Model: `rma.mv(yi = delta, V = vi, random = ~1|dataset_norm, test = "t", method = "REML")` on the 10 cells of Table 6.1.

Table D.23: Full cross-paper pool ($k = 10$).

Parameter	Value
k	10
Intercept ($\hat{\Delta}$)	+0.101
SE	0.145
$t(9)$	0.694
p-value	0.505
95% CI	$[-0.227, +0.429]$
σ^2 (dataset)	0.063
$Q(9)$	1055.3, $p < .0001$
I^2	99.1%

Interpretation. The full-pool intercept is positive (FM worse on average) but far from significant. The extreme heterogeneity ($I^2 = 99.1\%$) is driven by the three M5 WAPE cells ($\Delta = -0.54$ to -0.78), which are artefactual (§5.4.5), and the M5 WRMSSE cell ($\Delta = +0.41$), which goes in the opposite direction.

Appendix D.2. Excluding M5 WAPE ($k = 6$)

Same model specification, dropping the three M5 WAPE cells but retaining M5 WRMSSE.

Note: The $k = 6$ pool in this table excludes only the three M5 WAPE cells. The M5 WRMSSE cell ($k = 1$) is retained because WRMSSE does not suffer from the per-series denominator pathology. However, because the M5 WRMSSE cell is the only non-WAPE cell in the pool, the excl-M5 intercept is reported as the primary result to avoid conflating metric effects.

Interpretation. After removing the M5 WAPE artefact, the intercept is small and negative (FM slightly better) but not significant. Heterogeneity drops to zero: the six remaining cells (Favorita $\times$ 3 + Rohlik $\times$ 3) are homogeneous. This is the primary result reported in the Abstract and §9.

Table D.24: Excluding M5 WAPE ($k = 6$).

Parameter	Value
k	6
Intercept ($\hat{\Delta}$)	-0.044
SE	0.028
$t(5)$	-1.583
p-value	0.174
95% CI	$[-0.115, +0.027]$
σ^2 (dataset)	0.001
$Q(5)$	2.00, $p = 0.849$
I^2	0.0%

Appendix D.3. Within-Paper Only — Source B ($k = 9$)

Model: `rma.mv(yi = delta, V = vi, random = ~1|paper_id + ~1|paper_id/dataset_norm, test = "t", method = "REML")` on the 9 Source B paired cells (3 datasets × 3 horizons, paper_id = LOCAL_MAC_*). Variance proxy: $v_i = 1/n_{\text{valid}} = 0.01$.

Table D.25: Within-paper only — Source B ($k = 9$).

Parameter	Value
k	9
Intercept ($\hat{\Delta}$)	-0.245
SE	0.103
$t(8)$	-2.386
p-value	0.044
95% CI	$[-0.482, -0.008]$
σ_1^2 (paper_id)	0.043
σ_2^2 (paper/dataset)	0.043
$Q(8)$	76.00, $p < .0001$

Interpretation. The within-paper model reaches significance ($p = 0.044$) with a substantial negative intercept (FM better by ∼25 pp on average across all 9 cells). However, this result is dominated by the three M5 cells ($\Delta = -0.54$ to -0.78). After mentally excluding M5, the Favorita and Rohlik cells contribute $\Delta \in [-0.01, -0.09]$, consistent with the cross-paper excl-M5 finding. The within-paper model is not used as the primary result because (i) it has only one paper_id (no between-study variation) and (ii) the M5 WAPE artefact inflates the intercept.

Appendix D.4. Summary Across Sensitivity Variants

The three variants tell a consistent story: the apparent FM advantage in the full pool is a composition effect driven by the M5 WAPE artefact. Once M5 WAPE is excluded, the FM-vs-LGBM gap is small (-0.044 WAPE), not significant ($p = 0.174$), and homogeneous ($I^2 = 0\%$).

Table D.26: Summary of sensitivity variants.

Variant	k	$\hat{\Delta}$	95% CI	p
Full pool	10	+0.101	[−0.227, +0.429]	0.505
Excl-M5 WAPE	6	−0.044	[−0.115, +0.027]	0.174
Within-paper (B)	9	−0.245	[−0.482, −0.008]	0.044

Appendix E. Extraction Table (Abbreviated)

The full extraction table (`analysis/extraction_schema.csv`, 185 data rows + 1 header) records one row per (paper × model × dataset × metric) observation. The complete CSV is available in the supplementary materials. Table E.27 shows the key columns; columns omitted for space include `dataset_variant`, `n_series`, `series_length_median`, `frequency`, `fine_tuned`, `runtime_reported`, `gpu_hours`, `hardware`, and `notes`.

Appendix E.1. Column Definitions

Table E.27: Key columns in the extraction table.

Column	Description
`paper_id`	Unique row identifier: `{category}{seq}_{detail}`
`authors`	First author et al.
`year`	Publication year
`dataset`	Target dataset (M5, Favorita, Rohlik, GIFT-Eval, fev-bench, etc.)
`model_name`	Model as named by the paper
`model_family`	One of: foundation, deep_learning, ml_tree, statistical, ensemble
`zero_shot`	Yes / No — whether the model was applied without task-specific training
`horizon`	Forecast horizon (days) or "mixed" if aggregated
`eval_method`	rolling / fixed_origin / competition
`metric_name`	Reported metric (WAPE, WRMSSE, MASE, MAE, sMAPE, WQL, CRPS, win_rate_*)
`metric_value`	Reported value (numeric or qualitative rank)
`has_covariates`	Whether the model used exogenous covariates

Appendix E.2. Row Counts by Category

Appendix E.3. Source A vs Source B

- **Source A** (literature extraction): categories A–F (155 rows from 38 studies). These rows report accuracy numbers as published by the original authors.
- **Source B** (gap-filling experiments): OWN + LOCAL (30 rows). These rows are from our own experiments on consumer hardware (MacBook M-series MPS) and Azure ML (E4DS_V4), using the shared protocol of §5.1.

The extraction table is frozen at the version used for the meta-regression of §6. Any post-submission additions would require re-running the pooled models of §6.7.

Table E.28: Extraction table row counts by category.

Category	Description	Rows
A	Foundation model papers	48
B	Benchmark papers	42
C	M5 competition and retail-specific	38
D	Deep learning baselines	12
E	Surveys and meta-studies	7
F	Cost/efficiency analysis	8
OWN	Our LightGBM baselines (Source B)	12
LOCAL	Our FM experiments (Source B)	18
Total		**185**

Appendix F. Threats to Validity

Five threats specific to the Source B foundation model runs (§5.4), in decreasing order of magnitude:

1. **Extraction selection bias.** Papers that publish on M5 / Favorita / Rohlik are not a random sample of the FM literature — they are the subset that chose a retail task for their own evaluation. Papers that chose ETT / Weather / ECL (most of the LSF literature) are absent. The §6.6 heterogeneity diagnostic includes a `paper_chose_retail_as_primary_tas` binary flag to surface this.

2. **Source A vs Source B protocol drift.** Source A numbers are extracted as-reported; Source B numbers are run on our §5.1.3 protocol. When the two diverge on the same (model, dataset, horizon) cell, the divergence is a data point about protocol sensitivity, not about model quality. The three-model overlap between A and B is designed to quantify this drift, and §5.4.7 treats >5% gaps as findings.

3. **Local run is three small models only.** The four large models (Moirai 2.0, TimesFM 2.5, Chronos-2, Chronos-Bolt bigger variants) cannot be run locally and are therefore represented in Table 5.7 only by Source A. A reviewer who distrusts Source A on a specific large model cannot cross-check against our local run. We flag this as the single biggest residual risk in §7.

4. **Covariate asymmetry.** Four of the six FM families in Table 5.7 are univariate and cannot consume the §5.1.1 covariate sets. The LightGBM baselines in §5.2 use the full covariate sets. This biases the comparison toward LightGBM on covariate-driven datasets (M5: SNAP, prices; Favorita: oil, promotions) and is not correctable inside a zero-shot protocol. §6.2 treats covariate-aware vs univariate as a moderator.

5. **Top-30k Favorita cap propagates to the local run.** Same cap, same selection rule, same velocity bias as §5.1.1. Source A rows for Favorita include both fev-bench's top-54k subset and the full-Favorita subset where available; we flag `dataset_variant` per row so the §6 pool can restrict to comparable subsets.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

[Author Name]: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Data availability

The extraction table (`analysis/extraction_schema.csv`), all analysis scripts, and experiment logs are available at `https://github.com/MaciejR/large-scale-demand-forecasting-benchm`
The M5 dataset is publicly available via Kaggle; the Favorita dataset is publicly available via Kaggle; the Rohlik v2 dataset is available at `https://github.com/Rohlik/rohlik-orders-forecasting-`

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